

RESEARCH ARTICLE

How to Transform Resource-Intensive Industries Toward Regenerative Business Models: A Design Science Study

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ABSTRACT

Firms continue to rely on unsustainable practices and linear business models that push planetary boundaries to their limits. While the concept of the regenerative business model (RBM) has emerged to restore and enhance social–ecological systems, it remains conceptual and requires guidance on how to transform existing business models to prioritize planetary health and societal well-being. This study aims to develop a practical approach for firms to transform existing business models toward regeneration. Using a Design Science Research Methodology (DSRM), we designed and evaluated a three-step process template supported by transformative questions based on six design requirements (DRs) that guide a RBM transformation. Our findings show that regenerative transformation unfolds through three interconnected phases: re-grounding, re-wiring, and re-seizing business models. We advance theory by identifying the mechanisms of regenerative transformation, inform practice by offering a structured and actionable process, and contribute methodologically by demonstrating how design science can generate design knowledge for regeneration.

1 | Introduction

Nonsustainable business models have pushed planetary boundaries to their limits, threatening societal and ecological stability (Hahn and Tampe 2021; Konietzko et al. 2023; Steffen et al. 2015; Vlasov 2021). Sustainable business models (Boons and Lüdeke-Freund 2013; Lüdeke-Freund 2010) and circular business models (Civera et al. 2025; Geissdoerfer et al. 2020; Pieroni et al. 2020) have emerged as potential alternatives to nonsustainable business models. While they seek to reduce the environmental impact and social harm (Muslemani et al. 2021; Bocken et al. 2014), their core objective remains reactive harm reduction rather than active ecosystem regeneration. Therefore, scholars increasingly support adopting regenerative business models (RBMs) that actively contribute to enhancing social–ecological systems (Hahn and Tampe 2021; Hausdorf and Timm 2023; Konietzko et al. 2023). These business models take

a holistic and systemic perspective to sustainability in business, linking social–ecological systems with human activity while prioritizing planetary health and societal well-being (Das and Bocken 2024; Hahn and Tampe 2021; Konietzko et al. 2023). We adopt the definition of social–ecological systems from Folke et al. (2010), who define them as “integrated system[s] of ecosystems and human society with reciprocal feedbacks and interdependence” (Folke et al. 2010, 3), making it a central tenet for the coexistence of economic activity and planetary boundaries.

While extant literature has clarified *what* RBMs aim to achieve (Das and Bocken 2024; Konietzko et al. 2023; Hahn and Tampe 2021), scholars have yet to find answers to *how* to enact regeneration and transform existing business models toward actively contributing to the enhancement of social–ecological systems (Das and Bocken 2024; Hahn and Tampe 2021; Konietzko et al. 2023; Salonen et al. 2025).

Thus, the mechanisms that move firms from incremental to systemic sustainable transformation, remain underexplored (Das and Bocken 2024; Hahn and Tampe 2021; Hausdorf and Timm 2023). Although RBMs offer a promising response to the limitations of incremental sustainability approaches, research lacks a clear understanding of the mechanisms of how firms can undertake this shift (Das and Bocken 2024; Hahn and Tampe 2021; Hausdorf and Timm 2023). In particular, there is little knowledge about the organizational capabilities that firms need to move from linear or sustainable business models to RBMs (Das and Bocken 2024; Hahn and Tampe 2021). Scholarship therefore needs to articulate a design perspective to guide this transition in ways that practitioners can apply in organizational contexts (e.g., Salonen et al. 2025). Moreover, existing work often focuses on isolated regenerative capabilities, for example, regenerative supply chains (Bag and Rahman 2023; Bag et al. 2025; Gualandris et al. 2024), regenerative strategies (Salonen et al. 2025), or regenerative leadership (Paul et al. 2025), while firms, and particularly incumbents, require a holistic and integrated understanding of RBMs to advance systemic change (Hahn and Tampe 2021; Konietzko et al. 2023).

Incumbents from resource-intensive industries (RIIs) are a major contributor to the decline of the Earth's regenerative capacity (Wurlod and Noailly 2018; Zhu et al. 2021; Zimmerman et al. 2020). These industries are characterized by their high energy consumption, high levels of pollution (Wurlod and Noailly 2018; Zhu et al. 2021), and reliance on "take-make-use-dispose" processes (Geissdoerfer et al. 2017; Kirchherr et al. 2023). Prominent examples of RIIs include, for example, the chemical industry (De Faria et al. 2021; Martinez-Hernandez 2017), the construction industry (Andersson 2020), and the steel industry (Gonzalez Hernandez et al. 2018). Specifically, large incumbents in these industries encounter complex and multifaceted transformational barriers (Zimmerman et al. 2020). These include, for instance, extensive and complex supply chains that are difficult to align with sustainable practices (Schilling and Seuring 2024), legacy systems that are costly to replace (Ogunwole et al. 2023), and bureaucratic organizational structures that impede progress, all of which are obstacles to large-scale transformation (Soderstrom and Weber 2020). Consequently, large incumbents often focus on reactive and incremental product, process, or business model improvements rather than adopting a broader systemic perspective to drive a more holistic, sustainability transformation (Frishammar and Parida 2019; Liu et al. 2015; Zimmerman et al. 2020). Given these unresolved questions and the lack of clear guidance for organizational application, there is a need for design-oriented research that can translate the existing conceptual insights on RBMs into actionable pathways for transformation. To advance research and practice on the transformation toward RBMs, we ask: *How can incumbents in RIIs move beyond incremental sustainability-oriented change to achieve a regenerative business model transformation?*

To address this crucial theory–practice gap, which requires a method capable of producing practical and theoretically grounded design knowledge, we adopt the design science

research methodology (DSRM) from Peffers et al. (2007). This approach allows us to study how incumbents in RIIs can transform their existing business models into regenerative ones while also generating design knowledge that supports developing a practically useful solution artifact. For our research context, we focus on the chemical industry, commonly referred to as a major RII (Hart and Milstein 1999), as it plays a crucial role in societal functioning and exemplifies the challenges of transitioning from a linear, fossil-reliant, waste-intensive model to more sustainable and regenerative approaches (De Faria et al. 2021; Ghosh and Bakshi 2017; Martinez-Hernandez 2017). To deepen our understanding of regenerative transformation challenges in this industry, we conducted mediated semi-structured interviews (Braun and Clarke 2006; Gioia et al. 2013). As a result, we developed a three-step process template for an RBM transformation, as well as a complementary transformative questionnaire. These artifacts guide incumbents in reconfiguring their existing business model and initiate the transition toward regeneration. Finally, we demonstrated and evaluated the process template and transformative questionnaire in a workshop setting (Venable et al. 2016).

Our study offers theoretical, methodological, and practical contributions. For theory, we extend the literature on RBMs by highlighting the required organizational mechanisms that enable an RBM transformation (Das and Bocken 2024; Konietzko et al. 2023). Methodologically, by introducing DSRM (Dimov et al. 2023; Peffers et al. 2007; Tuunanen et al. 2024) to the RBM literature, we contribute to the growing body of work applying design science approaches to advanced sustainability challenges (e.g., Baldassarre et al. 2020; Remke and Müller 2024). This approach enables us to develop a prescriptive, theoretically grounded solution that addresses pressing issues in practice while advancing design knowledge in the domain of regenerative transformation for business. For practice, we provide managers of incumbents in RIIs with a useful solution for an RBM transformation. These artifacts serve as useful tools for enabling transformative pathways toward regeneration by assessing a firm's regenerative readiness, designing regenerative interventions, and evaluating the progress made.

2 | Theoretical Background

While regeneration has been discussed in other fields of research, it is a relatively new concept in management research. For example, in biology research, regeneration refers to the ability of a living entity to reconstruct itself after parts of it have been destroyed (Birbrair et al. 2013). In ecology research, regeneration describes the environment's and its living population's capacity to recover and renew after damage (Dietze and Clark 2008). Beyond biology and ecology research, regeneration is increasingly studied in fields such as agriculture (Rhodes 2017; Schreefel et al. 2020), tourism (Bellato et al. 2023; Dredge 2022), economics (Fath et al. 2019; Shannon et al. 2022), medicine (Mason and Dunnill 2008), engineering (Laurencin and Khan 2012), and urban studies (De Magalhães 2015; Fayed and Rashed 2020).

In management and sustainability research, regenerative businesses are defined as firms that “enhance, and thrive through the health of social–ecological systems in a co-evolutionary process” (Hahn and Tampe 2021, 456). These organizations align their business practices with the focus on enhancing planetary health and societal well-being (Das and Bocken 2024; Hausdorf and Timm 2023; Konietzko et al. 2023). In this context, regeneration is understood as the link between human activities and their engagement with social–ecological systems (Hahn and Tampe 2021; Muñoz and Branzei 2021; Vlasov 2021). As such, regenerative businesses aim at enhancing social–ecological systems and strengthening their response to external influences (Das and Bocken 2024; Muñoz and Branzei 2021). Social–ecological systems are defined as “integrated system[s] of ecosystems and human society with reciprocal feedbacks and interdependence” (Folke et al. 2010, 3). Regeneration, therefore, seeks to integrate and enhance both ecological (Bansal et al. 2025; Caldera et al. 2022; Paul et al. 2025) and social systems (Neisig 2025; Slawinski et al. 2021) by shifting the focus from harm reduction toward active preservation and enhancement (Hahn and Tampe 2021).

Conventionally, social–ecological systems have been regarded as sources of problems requiring solutions, rather than systems that can be actively improved to benefit their embedded actors (Dean and McMullen 2007; Muñoz and Cohen 2017). Since all actors are integrated within social–ecological systems, the complex, multivariable, and nonlinear challenges in constantly changing environments require unlearning and rethinking this status quo of viewing them as the source of problems (Ostrom 2007). This shift is essential for adopting regenerative practices (Buckton et al. 2023; Muñoz and Cohen 2017; Ostrom 2007).

A recent concept that has emerged in response to the need to embed regeneration at the core of organizations is the RBM. By adopting a holistic perspective on social–ecological systems, the RBM prioritizes planetary health and societal well-being across its value proposition, value creation and delivery, and value capture mechanisms (Hahn and Tampe 2021; Konietzko et al. 2023; Vlasov 2021). RBMs differ from linear, sustainable, and circular business models in their focus on actively contributing and generating a net-positive impact on social–ecological systems (Das and Bocken 2024; Konietzko et al. 2023; Shannon et al. 2022). In contrast, linear business models neglect the social and environmental impacts of business activities (Lüdeke-Freund 2010). Sustainable business models aim to reduce the adverse effects on natural ecosystems and local communities (Boons et al. 2013; Boons and Lüdeke-Freund 2013). Circular business models primarily focus on minimizing material and energy throughput while maximizing value creation through closed-loop economic systems (Geissdoerfer et al. 2017; Geissdoerfer et al. 2023).

While the RBM lacks a universally accepted definition, Konietzko et al. (2023) outline three key principles. First, these models recognize the interdependence of human societies and environmental ecosystems. Second, they embed societal well-being and planetary health into their value proposition. Third, they aim for a net-positive impact by actively contributing to the regeneration of the planet. Overall, RBMs adopt a holistic view of the relationship between social–ecological systems and

human activity (Hahn and Tampe 2021; Jovanović-Vujatović et al. 2024; Konietzko et al. 2023).

The value proposition of RBMs focuses on creating a net-positive impact on societal well-being and planetary health (Das and Bocken 2024; Hahn and Tampe 2021; Konietzko et al. 2023). This requires aligning the organization's values, meaning, and purpose with the goals of enhancing both planetary and societal health (Konietzko et al. 2023). Value creation and delivery of RBMs emphasizes regenerative leadership (a deep commitment to nature and society), co-partnerships with nature, and principles of justice and fairness (Hutchings and Storm 2019; Paul et al. 2025). This includes advocating for nature and animal rights, investing in natural capital, and articulating a clear organizational purpose (Hernández and Muñoz 2022; Konietzko et al. 2023). As Hernández and Muñoz (2022) emphasize, such practices are essential for embedding an eco-centric form of management that aligns with the regenerative purpose of the organization and shapes its decision-making processes. Finally, value capture of RBMs involves multi-capital accounting to ensure and account for net-positive impact (Hahn and Tampe 2021; Konietzko et al. 2023; Muñoz and Branzei 2021). Multi-capital accounting entails various forms of capital, including natural, cultural, social, financial, and human capital (Konietzko et al. 2023). Together, an RBM shifts individual-focused actions to a multilevel strategy that prioritizes systemic improvements over incremental gain (Du Plessis 2012; Hahn and Tampe 2021; Hausdorf and Timm 2023).

In practice, the adoption of RBMs remains limited. Despite growing scholarly interest, Das and Bocken (2024) only identify 84 ventures that meet the criteria for such models. While elements of regeneration become of increasing interest to research, such as regenerative supply chains (Bag and Rahman 2023; Bag et al. 2025; Gualandris et al. 2024), regenerative business strategies (Salonen et al. 2025), regenerative leadership (Paul et al. 2025), or regenerative knowledge capture (Rahman et al. 2024), RBMs are still in the early stages in both theory and practice. This is why the understanding of how to embed regeneration in organization and focusing on RBMs as one concept that combines multiple elements of business requires more conceptual and empirical grounding (Bansal et al. 2025; Branzei et al. 2017; Salonen et al. 2025).

3 | Method

3.1 | Data Collection and Analysis

The objective of this study is to identify pathways through which incumbents in RIIs can shift from incremental sustainability initiatives toward an RBM transformation. Accordingly, we examine how firms can move from marginal improvements to systemic approaches that actively contribute to social–ecological systems (Das and Bocken 2024; Folke et al. 2010). To capture this, we employed a DSRM (Dimov et al. 2023; Peffers et al. 2007; Tuunanen et al. 2024), which seeks to *prescriptively* develop practically useful artifacts that address theoretical and practical challenges while remaining consistent with the principles of scientific methods (Seckler et al. 2021). In doing so, DSRM bridges the gap between theory

and practice by informing the design of relevant solutions and fostering the development of theoretically grounded tools and frameworks for real-world application. The method has recently gained traction as a rigorous and increasingly adopted approach within research streams on sustainability, the circular economy, and business model innovation (e.g., Baldassarre et al. 2020; Müller and Remke 2025). The DSRM research process typically entails six coherent steps: problem definition, solution objectives, design and development, demonstration, evaluation, and communication (Peffer et al. 2007; Tuunanen et al. 2024). Figure 1 illustrates the six sequential stages of the DSRM process that guided our research design.

Following Eisenhardt et al. (2016), we apply an inductive research design that is well-suited for investigating grand sustainability challenges for unresolved problems. To conduct an in-depth problem analysis that informs the DSRM and extends our understanding beyond literature, we empirically collected primary qualitative data by using a semi-structured interview guide (see Appendix Interview Guides 1, 2, 3). To ensure a rigorous and inductive data collection procedure, we drafted the interview guide before familiarizing ourselves with the relevant literature, thus following Gioia et al. (2013). We employed purposive sampling in accordance with Etikan (2016) and selected interview participants for their engagement and expertise in RIIs in general and the chemical industry in particular. The final sample ($n = 23$) included the following: (1) chemists as the backbone of the regenerative transformation, to verify the technical feasibility, (2) business-driven chemistry innovators (i.e., entrepreneurs) to clarify key barriers to innovation in RIIs and gain further insights into practical challenges, as well as (3) sustainability and regeneration experts to supplement and evaluate our findings. We chose these interviewee profiles because they cover all relevant aspects necessary to fully understand the dynamics of RIIs and requirements for integrating regeneration into incumbents current business models. This approach ensured a holistic, multidimensional identification of relevant themes,

which helped to define relevant design requirements (DRs) for designing a solution. Between November 2023 and April 2024, we conducted 23 semi-structured interviews. With participants' informed consent, the interviews were recorded and conducted via Zoom, lasting an average of 45 min (see Table 1 for overview of interview participants).

In addition to the interview data, we collected supplementary material from the interviewees' firms to validate and enrich our data collection process and to ensure scientific rigor, commonly referred to as data triangulation (Carter et al. 2014). Specifically, we reviewed sustainability reports, examined references to regeneration on corporate websites, and analyzed content from social media accounts. For data analysis, we followed an inductive qualitative approach and employed open, axial, and sequential coding (Braun and Clarke 2006; Gioia et al. 2013; Strauss and Corbin 2003). From the initial coding, we derived first-order concepts, second-order themes, and aggregate dimensions (Braun and Clarke 2006; Gioia et al. 2013). For example, the verbatim statement "If you are sustainable, then what? Nature is not really going to say thank you." was coded as the first-order concept "Skepticism toward Sustainability." This evolved into the second-order theme "Transforming Mindsets, Norms, and Economic Paradigms," which directly informed the formulation of a DR. This was then integrated into the aggregate dimension "Culture and Education," representing a core component of our solution artifact. All authors were involved in the coding process and iteratively compared results among each other to strengthen inter-coder agreement (see Figure 2 of our data structure).

Following the six-step DSRM sequential process by Peffer et al. (2007), we first defined the problem in analyzing extant theoretical knowledge and the research context underlying our study. The primary issue we identified is the persistent linearity in business models of RIIs, such as the chemical industry, and the resulting negative social and environmental

Nominal process sequence

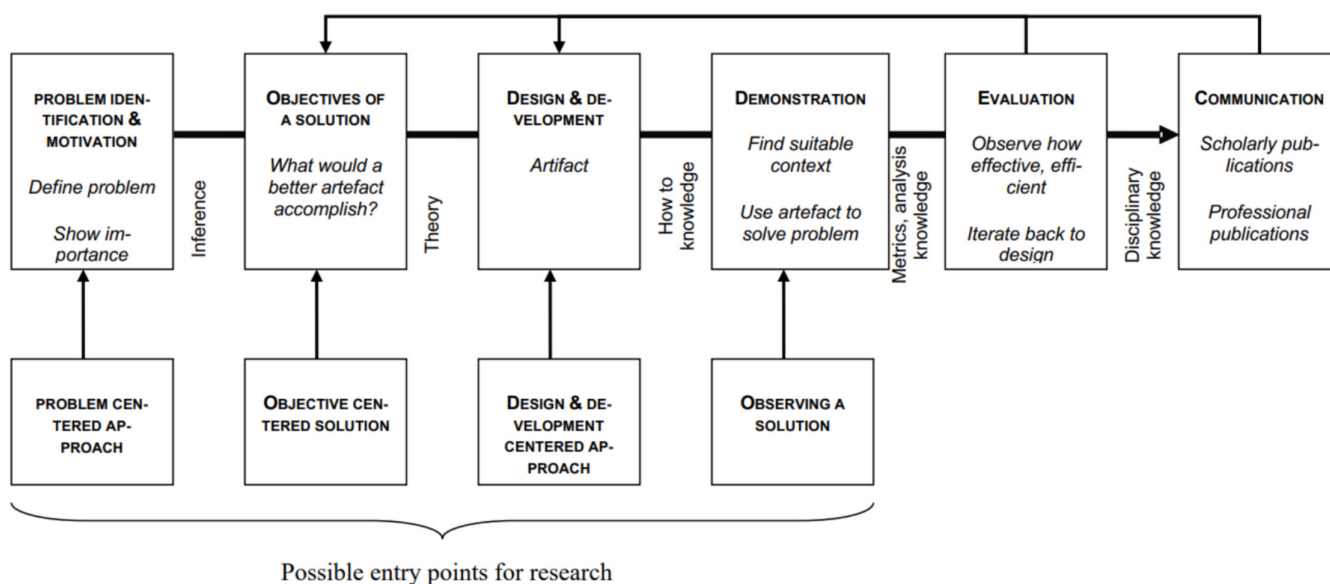


FIGURE 1 | Representation of design science research methodology process sequence from Peffer et al. 2007.

TABLE 1 | List of interview participants in the exploratory phase.

Participant no.	Role	Target group number	Geographical location
1	Sustainability and Regeneration Consultant	3	Germany
2	Entrepreneur	2	Germany
3	Chemical Engineer	1	Germany
4	Chemical Engineer	1	Canada
5	Chemist, Entrepreneur	1, 2	France
6	Chemist	1	Germany
7	Entrepreneur	2	France
8	Chemistry Expert	1	France
9	Chemist, Eco-Industrial Parks Expert	1, 2	Ukraine
10	Chemical Project Manager	1, 2	France
11	Entrepreneur	1, 2	Germany
12	Chemist	1	France
13	Business Developer	2	Germany
14	Polymer Chemist	1	Germany
15	Chemistry	1	France
16	Entrepreneur	3	Germany
17a	Entrepreneur, Pioneer	3	France
17b ¹	Entrepreneur, Pioneer	3	France
18	Chemist, Partnership Expert	1, 2	Germany
19	Entrepreneur	3	Germany
20	Sustainability and Regeneration Consultant	3	Canada
21	Biochemist	1	Germany
22	Chemist, CVC Manager	1, 2	Germany

¹Sessions 17a and 17b represent two sequential group sessions facilitated by the same lead entrepreneur. Session 17a focused on conceptual dimensions of regeneration, whereas Session 17b centered on practical application through a product-level workshop with multiple experts.

consequences (e.g., Zimmerman et al. 2020). For the second step of the DSRM process, we follow the guidance provided by Hevner et al. (2004) and the blueprint by Trunk et al. (2025) for deriving objectives of a solution. We translated our data into actionable DRs. To enhance the robustness and rigor of the DRs, we triangulated the interview data with supplementary materials, including firm reports, publications, and social media mentions related to regeneration. Conceptually, we guide the development of the DRs following Van Aken and Berends' (2018) framework for organizational problem-solving. The framework consists of four key types of DRs: functional requirements, user requirements, boundary conditions, and design restriction. Using this framework, we systematically derived DRs to inform the subsequent design and development of a solution. In the third step, we developed a solution artifact which serves as a valuable solution to the identified problem, hence the RBM transformation process template. In the fourth step of the DSRM process, we demonstrated and evaluated the solution in a workshop setting as proposed by Venable et al. (2016). The workshop lasted 2 hours and was conducted in a mediated and synchronous online

environment via the platform Zoom. During the workshop, we guided participants through the proposed solution and sequentially collected feedback on the different elements of the solution. In the fifth and sixth steps of the DSRM process, we refined the solution based on the feedback received throughout the workshop and communicate the final solution through the publication of this study. Table 2 provides an overview of the seven workshop participants who evaluated our process template, including their roles, expertise, and geographical locations.

3.2 | Research Context

For the research context, we focus on the chemical industry as one of the most significant RIIs (Martinez-Hernandez 2017). This industry is crucial to the global economy, accounting for over 95% of manufactured goods (De Faria et al. 2021; Martinez-Hernandez 2017). It plays an essential role in our daily lives and the functioning of society, with many sectors relying on chemical products and their byproducts

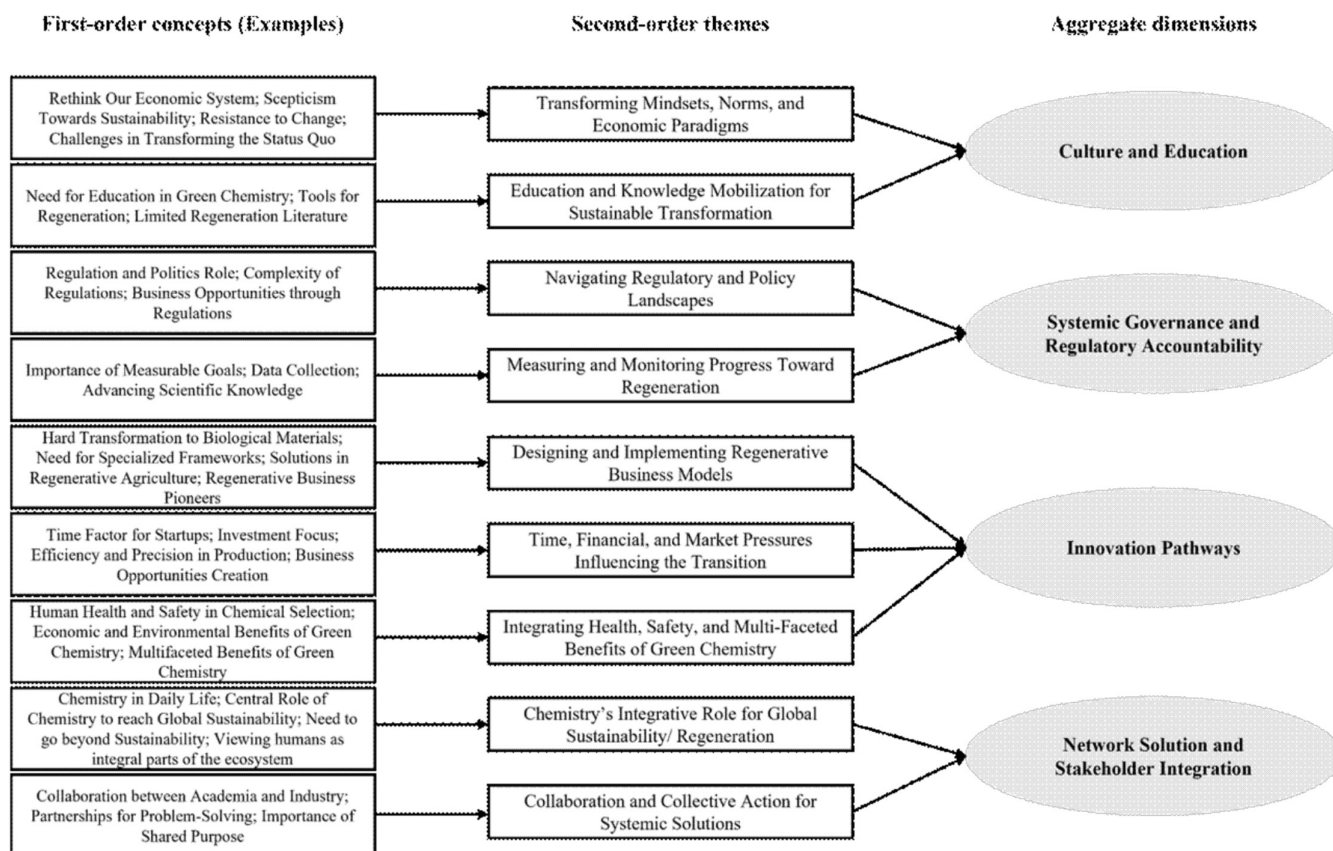


FIGURE 2 | Data structure (Source: Structure based on Gioia et al. 2013).

TABLE 2 | List of participants in the evaluation phase.

Participant no.	Role	Target group number	Geographical location
1	CEO	2	Germany
2	Entrepreneur	3	Netherlands
3	Chemist, Entrepreneur	1, 2	France
4	Entrepreneur	3	Portugal
5	Entrepreneur	2	Germany
6	Regeneration Expert	3	France
7	Chemist	1	France

(Martinez-Hernandez 2017; Zimmerman et al. 2020). The industry's predominant focus on linear business models, which are characterized by a resource-to-waste flow, relies heavily on fossil resources, harmful chemicals, and energy-intensive processes, resulting in significant amounts of non-recycled waste (Zimmerman et al. 2020). This reliance presents a substantial challenge to decarbonizing and reducing its environmental impact (De Faria et al. 2021; Ghosh and Bakshi 2017). Proposals to address these issues include utilizing renewable

resources, waste valorization, as well as promoting reuse and recycling to reduce environmental impacts (Van Geem et al. 2019; Zhu et al. 2021). While these proposals are certainly valuable from a sustainability perspective, they do not yet correspond with shifting the focus from harm reduction toward active preservation and enhancement; the fundamentals of regeneration (Hahn and Tampe 2021). Given the chemical industry's importance, Zimmerman et al. (2020) call for a thorough reassessment of "the character, nature and production processes of synthetic chemicals [...] for a sustainable civilization" (p. 397). Thus, aligned with the fundamentals of regeneration, scholars call for maintaining or improving chemical performance at the product, process, and firm level while moving beyond minimizing negative impacts (Matlin et al. 2016; Sheldon 2017; Zimmerman et al. 2020). This thorough integration of the chemical industry into our everyday lives and the theoretical demand to rethink the prevalence of nonsustainable business models make this research context particularly compelling.

4 | Findings

After defining the problem through examining extant literature and laying out the research context, in the second step of the DSRM process, we derive actionable DRs that guided the development of our solution artifact. The DRs were derived from the categories that arose from our data analysis procedure (Braun and Clarke 2006; Gioia et al. 2013).

4.1 | Establishment of DRs

Through our qualitative, inductive data collection, we find that integrating regeneration into the linear value chain of the chemical industry represents a radical mindset shift that is likely to encounter strong resistance (Interviews 13, 20). Regeneration, as an emerging sustainability concept, demands a move away from fossil fuel dependency toward more natural processes. However, the chemical industry has traditionally been resistant to such change (Interviews 18, 20). Although theory and practice have advanced in improving the industry's sustainability, skepticism remains widespread. As Interviewee 18 notes, "If you are sustainable, then what? Nature is not really going to say thank you."

We find that this resistance often reflects a generational mindset conflict, adding nonrational challenges to the already complex technical nature of these transformations (Interviews 13, 16, 20). Interviewee 16 emphasized that "most of the time, incumbent organizations are having a hard time to change." The perception of immutability in processes, and its reflection across the organization, shows that meaningful transformation requires rethinking core values and beliefs. These are crucial, as they shape operational decision-making in practice. This aligns with public information from Interviewees 13 and 16 firms on shifting mental models toward regeneration as "change takes time and is often a painful process. The beginning of change involves loss and triggers denial, anxiety, shock, etc." Consequently, this shift requires acknowledging the industry's impact on social-ecological systems (Interview 13). However, the broader system in which the chemical industry operates does not favor a radical transition toward regenerative thinking. Interviewee 16 illustrated this tension, calling for a move "from an economy that is monetizing negative externalities [...] toward an economy that is monetizing positive externalities." As a result, global ecosystems are often perceived as detached from human agency (Interview 13). This limited view of human influence on social-ecological systems reveals the cognitive and cultural barriers for adopting regenerative concepts. To overcome these barriers, organizations must foster a learning culture that educates employees and reestablishes a symbiotic relationship between humans and ecosystems. As Interviewee 20 described, organizations need to begin by "viewing humans as integral parts of the ecosystem rather than superior entities" and by "moving from human-centric to eco-centric design philosophies." Based on this, we define DR1:

The solution must foster regenerative thinking and eco-centric practices to establish a foundational shift in practitioners' mindsets across all organizational levels.

To foster a mindset shift that enables organizations to regard themselves as active agents in regenerating social-ecological systems, it requires corresponding changes in a firm's culture and education. Despite advancements in RIIs, and particularly in the chemical industry (e.g., green chemistry initiatives), there remains a critical need for educational programs and tools that build regenerative capabilities across all organizational levels. Regeneration aims at enabling system-level change and therefore

hinges on learning and knowledge development to enact these shifts. Interviewees 1, 2, 9, and 14 consistently emphasized the central role of education in addressing complex global challenges within their respective industries. As Interviewee 2 noted, "there's a lot of work to do in terms of education," while Interviewee 9 highlights the transformative potential of this effort: "To educate future decision makers [...] and to promote research on sustainable chemistry, to make it cheaper." We compared this to firm information on developing regenerative leadership skills "[...] through their interactions with other stakeholders, each person contributes to the 'regeneration' of the other. For example, by listening empathetically, allowing each person to be understood in their uniqueness, including their vulnerabilities, and to develop their own potential."

Although the principles of regeneration have been explored in disciplines within and beyond management, their integration into organizational contexts requires new educational approaches that balance foundational understanding with the multidimensional challenges faced by RIIs. Unlike traditional approaches that often compartmentalize environmental and human aspects, regeneration emphasizes the interconnectedness of all elements within the ecosystem (Interviewee 17a). This is why regenerative principles require innovative educational solutions to address multidimensional challenges across various fields. Enabling the generation of new regenerative knowledge requires learning new skills that were previously not needed and to unlearn practices that solely focus on harm reduction or disregard social-ecological harm. Based on this, we propose DR2:

The solution must enable practitioners to build regenerative capabilities by fostering learning that supports the design and implementation of regenerative solutions.

As the literature on regenerative practices for firms is still in its infancy, practitioners express a strong need for actionable guidance to develop the educational materials outlined in DR2. The absence of practical business model references poses a major barrier to advancing transformative regenerative change. For the chemical industry to transition toward regeneration, our interviewees repeatedly emphasized the importance of developing proof-of-concept initiatives and pilot projects to demonstrate feasibility and to drive broader adoption (Interviews 2, 18, and 19). However, the scarcity of practical tools and conceptual frameworks continues to impede the initiation of such projects. Consequently, research on agricultural issues within the broader field of regeneration often serves as an initial point of reference, as it provides tangible examples that help practitioners, "create the first solutions, and show proof that this is actually working" (Interview 16). While these cases offer valuable inspiration, translating regenerative concepts from agriculture or biology to RIIs introduces additional layers of complexity. Moreover, existing sustainable and circular business model design tools frequently fail to capture the specific requirements and constraints of the chemical sector. Interviewee 11, for instance, described the limitations of applying a traditional business model canvas to a chemical business context. In addition, interview participants consistently emphasized the urgency of

moving beyond harm reduction toward business models that actively regenerate social–ecological systems (Interviews 12, 14, 17a). Considering these insights, we derive DR3:

The solution must provide practical guidance and reference points of regenerative practices that enable firms to design, test, and scale regenerative initiatives to gradually transform the business model.

Chemistry is characterized as the science of material transformations and draws inspiration from the natural environment. Drawing on natural processes is widely identified in interviews as a key factor in developing products and processes. As a result, both chemical lab research and business activities are increasingly leveraging bio-resources as drivers for a sustainable environmental transformation (Interviews 7, 8). At the same time, different stakeholders in the chemical industry face diverse challenges, needs, and objectives, requiring a customizable and holistic approach to address the entire value chain. As the complexity of the chemical industry demands different foci at different stages of its value chain, integrating regeneration poses an issue of competing interests. According to Interviewee 2, a potential solution would require catering to different stakeholders and actors needs. As such, entrepreneurs and chemical business innovators would need to involve “regenerative design guidelines [...] to focus on circularity and adaptability.” On the contrary, this focus on adaptability of the solution to various needs would not work for chemists in industrial production. Here, Interviewee 3 emphasizes that chemists especially prioritize efficiency and accuracy at work. These differing priorities highlight the complex, multidimensional challenges faced by the industry and its stakeholders. This is also highlighted by the online newsletter of this firm, as regeneration requires a co-creation and practical references to start: “Thus, transforming your business model requires innovation and the launch of products and services resulting from regenerative practices.” To address these, and to accommodate the diverse perspectives and needs along the value chain, we derive DR4:

The solution must enable the customization of regenerative approaches to diverse, complex, and multi-dimensional priorities and challenges of RIIs.

Different actors within the chemical industry have distinct needs that must be reflected in a useful tool. Throughout our interviews, participants consistently highlighted the importance of aligning practically useful business model solutions with regulatory frameworks (Interviews 1, 16, 18). While regulatory requirements are often perceived as burdens that constrain the industry's flexibility, Interviewee 1 pointed out that regulatory and political factors are, in fact, crucial drivers of sustainability initiatives, shaping both strategic and operational decision-making. Regulatory sustainability initiatives such as ESG have contributed to the unification of reporting metrics, enhancing comparability across actors within the chemical industry. For a regenerative approach to achieve similar adoption and effectiveness, it requires standardized and unified reporting frameworks that ensure consistent evaluation among stakeholders (Interviews 1, 18). Establishing such metrics for regeneration would improve transparency and comparability across firms,

helping to identify actors engaging in degenerative practices while incentivizing others to maximize regenerative outcomes through shared standards.

In summary, while regulations may introduce challenges, they also create “business opportunities for companies to adapt” (Interview 18). Across our interviews, regulations were consistently viewed as catalysts for innovation and the development of intellectual property (Interviews 2, 5, 16). Therefore, a suitable tool should not only guide the integration of standardized regenerative measures but also advocate for cross-industry comparability and stimulate innovation within a clear and supportive regulatory framework, which leads to DR5:

The solution must account for existing sustainability regulations in RIIs and advocate for standardized, comparable regenerative metrics across the value chain.

Lastly, developing a solution to transform existing business models into regenerative ones must account for the scale and intensity of transformation required within the chemical industry. This involves several interrelated dimensions, for example, geographical factors that impact the structure of regional supply chains and resource availability (Interviews 2, 13, 18). As regeneration focuses on positively contributing to the social–ecological systems it is embedded in, the location of chemical production sites constrains the extent and nature of what can be produced. It is therefore essential to align the scale of transformation with the characteristics of the local natural environment (Interviews 2, 13, 18). Beyond geography, time has emerged as another relevant aspect in our data collection process. Time shapes the development of sustainable chemical products and processes, influencing research efforts across academia, laboratory experimentation, and industrial production. The transition from research to large-scale implementation can take several years, as regenerative solutions must meet stringent safety and durability standards. As a result, addressing time-related considerations is essential for the survival and success of businesses in the chemical industry and other RIIs (Interviews 2, 4, 10, 14, 15).

Finally, the transformation of an entire industry cannot be driven by isolated actors, it requires collective effort and collaboration. Cross-sector partnerships foster shared learning, resource pooling, and coordinated action, enabling more efficient problem-solving and generating both financial and ecological benefits. As several interviewees emphasized, shared goals and mutual reinforcement among actors are vital for achieving systemic and lasting regenerative impact (Interviews 11, 15, 18). This is in line with public information to promote a coalition of change agents “Achieving the ecological transition requires a coalition of change agents.” Thus, this is how we define DR6 (Table 3):

The solution must account for the spatial, temporal, and collaborative dimensions of transformation in RIIs.

TABLE 3 | Design requirements (DR).

DR	Description	Type (Van Aken and Berends 2018)
DR1	The solution must foster regenerative thinking and eco-centric practices to establish a foundational shift in practitioners' mindsets across all organizational levels.	Boundary condition
DR2	The solution must enable practitioners to build regenerative capabilities by fostering learning that support the design and implementation of regenerative solutions.	User requirement
DR3	The solution must provide practical guidance and references points of regenerative practices that enables to design, test, and scale regenerative initiatives to gradually transform the business model.	User requirement
DR4	The solution must enable the customization of regenerative approaches to diverse, complex, and multidimensional priorities and challenges of RIIs.	User requirements
DR5	The solution must account for existing sustainability regulations in RIIs and advocate for standardized, comparable regenerative metrics across the value chain.	Functional requirements
DR6	The solution must account for the spatial, temporal, and collaborative dimensions of transformation in RIIs.	Design restriction

4.2 | Artifact Design and Development

Based on the identified DRs, we developed a three-step process template and complementary questionnaire to guide the transformation from non-regenerative (linear, sustainable, or circular) toward RBMs. The artifact shall serve as a practically useful solution for incumbents in RIIs, particularly the chemical sector, to transform their business models to actively enhance social-ecological systems. Thus, we provide a structured approach to assess regenerative transformation readiness, design transformation pathways, and evaluate progress. In this structured approach, the DRs served as core attributes that guided each iteration of artifact development. Based on our inductive data analysis, we identified four aggregate dimensions, *Culture and Education*, *Systemic Governance and Regulatory Accountability*, *Innovation Pathways*, and *Networked Solutions*, which represent the key transformative areas of an RBM development (Braun and Clarke 2006; Gioia et al. 2013). The process combined insights from the literature which were merged with empirical data in a collaborative, iterative approach by the authors. Brainstorming sessions were used to develop and refine the process structure, followed by multiple feedback loops.

4.3 | Conceptual Foundations and Design of the Process Template

The conceptual foundation of our process template is rooted in Lewin's freeze-unfreeze-refreeze change management model (Lewin 1947) and is based on the process blueprint developed by Müller and Remke (2025) for circular systems design. We integrate the insights from these change-oriented process models into the design of our process template. As such, both Lewin (1947) and Müller and Remke (2025) generate a new and revised state of organizational progress or existence by assessing the current state of the organization (freeze), designing an

intervention that guides the transformation (unfreeze), and an evaluated integration of these to improve on the changes (refreeze). Müller and Remke (2025) Circular Systems Sandbox guides firms through a transformation process consisting of specific phases and step that supports the development of a multilayered circular system design. We used this framework as a visual and procedural scaffold to inform the development of our process model. Following Seckler et al. (2021) in their guidance for planning, conducting, and assessing design science research, we position this study as an *improvement study* within DSRM (e.g., Gregor and Hevner 2013). Specifically, we advance existing design knowledge, such as the Circular Systems Sandbox, by refining and extending it to address the requirements of RBMs. Figure 3 presents our final three-step process template for an RBM transformation, visually depicting the assessment, intervention, and evaluation stages.

4.4 | The Process Template

The final artifact takes the form of a three-step process template to guide the transformation of existing business models toward regeneration. This process template draws on the identified DRs for a structured transformation journey. The process template is split into three distinct stages. In the first step, firms need to assess their readiness for a regenerative transformation (freeze). In the second step, firms design and implement regenerative interventions to transform existing business model processes (unfreeze). Lastly, in the third step, these interventions are evaluated by their outcomes and iterated upon (refreeze). We complement the process template with a targeted, diagnostic questionnaire that aims at generating concrete regenerative transformation initiatives. This ensures that firms can identify leverage points, initiate concrete actions, and monitor the impact of their transformation efforts.

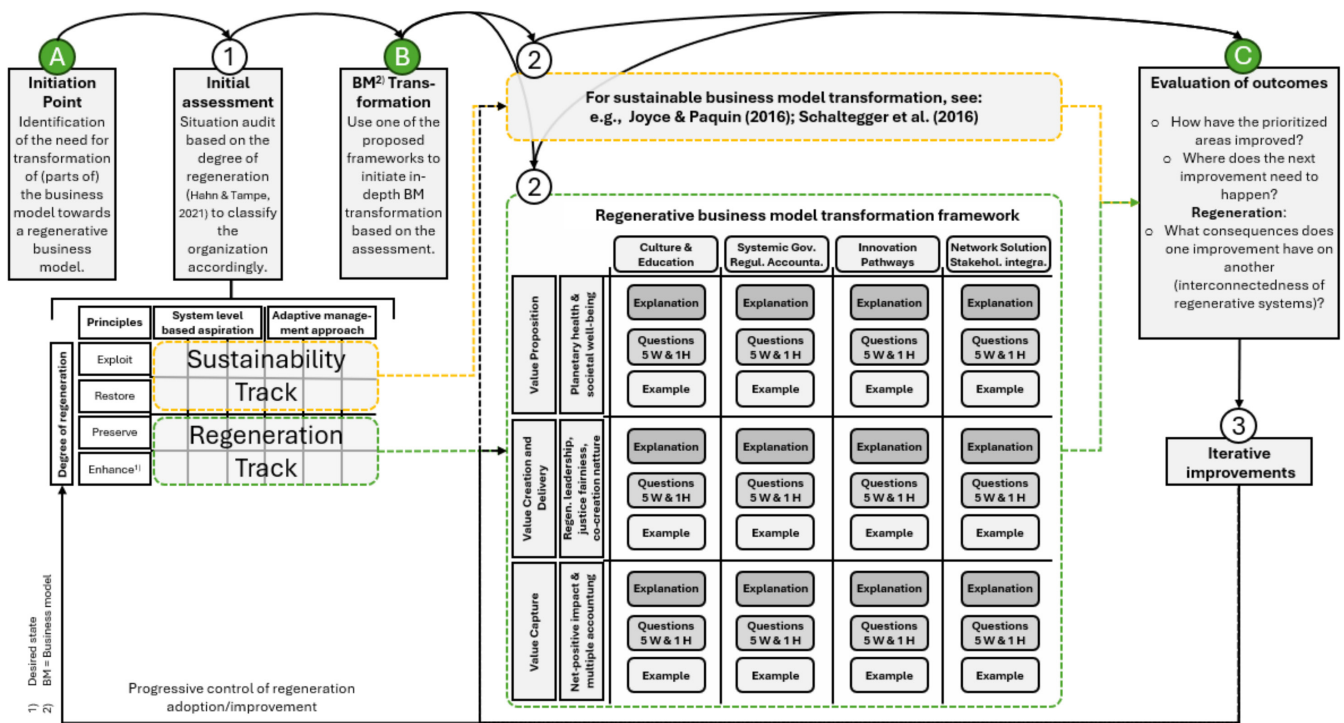


FIGURE 3 | Process template tool—regenerative business model transformation.

4.5 | Readiness Assessment for Transformation Pathway

The first step of our process template builds the foundation for evaluating a firm's readiness to transform to an RBM. This step is essential for identifying both the need for a regenerative transformation and the organization's capacity to realize it. To assess the readiness for a regenerative transformation, firms conduct a self-assessment using Hahn and Tampe's (2021) *Degrees of Regeneration* framework. This framework classifies business practices along a continuum of four progressive stages ranging from non-regenerative to increasingly regenerative practices (*exploit*, *restore*, *preserve*, and *enhance*). These stages indicate the degree to which business activities either impose harm (*exploit*) or contribute positively to social-ecological systems (*preserve*, *enhance*) (see Figure 3).

We integrate this framework to classify current levels of regenerative practices, but at the same time, it serves to evaluate a firm's understanding of its interrelated embeddedness in social-ecological systems and ability to manage complexity across system levels (Hahn and Tampe 2021). This helps firms understand not only where they currently stand on regeneration but also how they can manage change and the complexity of regeneration in practice. This dual assessment provides a comprehensive picture of both position and potential for transformation.

Building on this foundation and to guide an RBM transformation, we divide the framework's four degrees of regeneration into two transformation tracks: (1) the sustainability track and (2) the regeneration track. Firms that self-assess their current business model within the *exploit* or *restore* categories are directed to first transform their business models for sustainability. For this stage, we recommend long-standing business

model frameworks, such as the Triple-Layered Business Model Canvas by Joyce and Paquin (2016) or taking examples from the co-evolutionary sustainable business model transformation by Schaltegger et al. (2016). To avoid overwhelming firms with advanced regeneration requirements, they first progress through the sustainability track before advancing to the regeneration track.

4.6 | Design and Implementation of Regeneration Intervention

The second step of our process template focuses on developing initiatives to transform the key areas of its business model. If firms self-assess for the regeneration transformation track, this step translates the insights from the assessment phase into specific actions that advance regenerative transformation across all business model dimensions. The framework builds on Konietzko et al.'s (2023) RBM principles. We integrate this with the four essential areas identified during our problem analysis: Culture and Education, Systemic Governance and Regulatory Accountability, Innovation Pathways, and Networked Solutions. Together, these quadrants constitute an integrated framework that highlight the crucial multidimensional nature of an RBM transformation and provides practical guidance for redesigning business model areas toward regeneration.

At the core of this step is a structured set of guiding questions (referred to as complementary questionnaire) that require reflection of current practices to initiate regenerative transformation. For each dimension, firms are provided with (1) an explanation of the desired state of regeneration (2) six diagnostic, targeted questions that address and answer the *what*, *why*, *where*, *how*, *when*, and *who* of an RBM transformation, and (3) an illustrative

TABLE 4 | Regenerative business model transformation framework value proposition.

	Culture and education	Systemic governance and regulatory accountability	Innovation pathways	Network solutions and stakeholder integration
Explanation	We foster a healthy and meaningful workplace culture that aligns with the organization's core values. We create intrinsic motivation to contribute to the organization's goals and broader societal values.	We advocate for systemic governance and regulatory accountability that prioritizes planetary health and societal well-being. By aligning our regenerative values with adaptive policies and collaborative metrics, we foster accountability, resilience, and outcomes that reflect ecological and social interdependence.	We focus on innovation pathways that create regenerative products, services, and processes benefiting both people and the planet. By integrating ecological knowledge, green chemistry, and nature-inspired solutions into our R&D, we align design principles with social-ecological systems interdependence.	We build collaborative networks and integrate diverse stakeholders to address systemic challenges and foster regenerative solutions. By establishing cross-sector partnerships, co-creation platforms, and knowledge-sharing frameworks, we align efforts with ecological interdependence and enhance community well-being.
Transformative questions	What: What existing norms, mindsets, or understandings of economics need to be challenged or unlearned to fully recognize the interconnectedness of social-ecological systems? Why: Why do we as an organization need to demonstrate the interconnectedness of societal well-being and planetary health? Why do we want to achieve lasting regenerative outcomes for our organization? Where: Where in our current educational programs, training sessions, or internal communications can we integrate lessons on social-ecological interdependence? Where can we emphasize regeneration in our culture? How: How can we design learning experiences that help participants internalize and apply the principle of regeneration in their daily decision-making? When: When and in what frequency should we introduce educational learning that focuses on regenerative business practices so that our employees maximize their impact on shaping long-term thinking and behavior? Who: Who are the influential educators, thought leaders, and cultural champions within and outside our organization that can foster an understanding of social-ecological systems interdependence?	What: What governance frameworks or accountability mechanisms can we develop to ensure decisions reflect their impact on social-ecological systems? Why: Why is integrating social-ecological systems interdependencies critical to govern regenerative outcomes and long-term resilience? Where: Where can we integrate multi-stakeholder input into monitoring processes to ensure local ecosystems and communities and increase accountability and systemic impact? How: How can we re/design governance structures and encourage cross-industry learning and continuous adaptation to the evolving ecological complexities? When: When should we proactively advocate to update policies and metrics to stay aligned with ecological and societal standards? Who: Who (which structures) must be represented in governance and oversight structures to ensure our measurement systems reflect ecological and social interdependencies?	What: What innovative design principles can we adopt to ensure that our value proposition is reflected within our products, services, and supply chains and promote the enhancement of social-ecological systems? Why: Why is this interdependence essential for increasing the identification of innovation opportunities that benefit both communities and ecosystems? Where: Where in our research and development pipeline can we embed ecological and systems thinking to drive transformative, regenerative innovation at scale? How: How can we use green chemistry, biomimicry, and other nature-inspired approaches to develop solutions that strengthen ecological resilience, societal well-being, and systemic regeneration? When: When and how frequently should we iterate and adapt our conceptual product or service designs or ideation processes to incorporate new ecological insights, stakeholder feedback, and scientific advancements? Who: Who within our innovation teams and partner networks can lead and guide the creation process of identifying and driving the redesign of our value proposition for regenerative solutions?	What: What collaborative frameworks or alliances can we establish to create shared value that prioritizes human health, societal well-being, and the regeneration of natural systems? Why: Why is it necessary to create value beyond our organization frame to effectively build network solutions that collectively enhance local and global regeneration? Where: Where should we establish cross-sector partnerships (e.g., between research institutions, businesses, and local communities) to address systemic challenges? How: How can we design knowledge-sharing and co-creation platforms that empower participants to collaboratively innovate, learn, and contribute to solutions that promote both human and planetary health? When: When should stakeholders align on priorities, measure collective impact, and adapt strategies based on evolving ecological and societal needs? Who: Who among our stakeholders can bridge sectoral gaps to co-create holistic, regenerative solutions?

(Continues)

TABLE 4 | (Continued)

	Culture and education	Systemic governance and regulatory accountability	Innovation pathways	Network solutions and stakeholder integration
Example	Phipps Conservatory and Botanical Gardens built the Center for Sustainable Landscapes, a net-positive energy and net-zero water building that provides a healthy and inspiring work environment (Phipps Conservatory and Botanical Gardens 2024).	Interface, a global carpet tile manufacturer, has established processes to manage time, resources, and waste through its “Mission Zero” initiative, aimed at eliminating its environmental impact by 2020. They optimize natural capital regeneration by using recycled materials and sustainable practices, such as closed-loop manufacturing (Fedrizzi et al. 2020).	Notpla creates biodegradable food packaging from seaweed, addressing environmental challenges by providing an alternative to plastic and supporting seaweed farming communities (Doan 2023).	Patagonia’s founder Yvon Chouinard declared, “Earth is now our only shareholder”, restructuring the company to dedicate profits to ecosystem conservation and regeneration (Patagonia 2021).

example from an organization that has successfully implemented principles of an RBM. In this way, we integrated the identified DRs, which allow firms from RIIs to contextualize regeneration within their specific organizational environment, identify transformation barriers, and develop actionable pathways for an RBM change.

To exemplify the framework’s logic, we highlight the *Value Proposition* as well as the *Culture and Education* quadrant. Culture and Education reflects the principle that regenerative transformation begins with a shift in organizational values and thinking. This area encompasses changes in mindset, the development of new skills, and the creation of learning infrastructures that sustain long-term transformation. Regeneration requires re-grounding the firm’s value proposition by aligning its values with the social and ecological well-being of its environment. Building on the “enhance” state of Hahn and Tampe’s (2021) Degrees of Regeneration framework, firms are encouraged to foster a meaningful workplace culture, cultivate intrinsic motivation, and strengthen alignment between organizational goals and broader societal values.

This complementary questionnaire then helps firms generate initiatives for a regenerative transformation that *re-wires* existing approaches to business practices. They prompt reflection on *what* aspects of the organization need transformation, *why* these changes matter, *where* in the firm they should occur, *how* they can be implemented, *when* they should take place, and *who* will take responsibility. For example, firms may ask and answer how they can integrate regeneration into internal training programs or how to design learning experiences that help employees apply regenerative principles in everyday decision-making (see Tables 4, 5, 6 for full questionnaire).

We complement these questions with practical “best practice” examples to illustrate what an RBM transformation can ideally look like in practice. One such case is the Phipps Conservatory, which created a “Center for Sustainable Landscapes” achieving net-positive energy and net-zero water use. This initiative enhances environmental outcomes as well as fosters a workplace culture centered on planetary health and societal well-being (Phipps Conservatory 2024). Such examples provide specific reference points for firms seeking to *re-seize* their value capture through regeneration.

Tables 4, 5, and 6 present the complete transformative questionnaire organized around the three business model dimensions. Table 4 addresses value proposition, Table 5 focuses on value creation and delivery, and Table 6 covers value capture. Each table provides diagnostic questions across the four transformation areas (Culture and Education, Systemic Governance and Regulatory Accountability, Innovation Pathways, and Network Solutions) along with illustrative examples from organizations that have successfully implemented regenerative principles.

The remaining dimensions follow the same logic. *Systemic Governance and Regulatory Accountability* (Dimension 2) focuses on coherent governance structures, policy alignment, and standardized metrics to ensure accountability and traceability of regenerative goals. *Innovation Pathways* (Dimension

TABLE 5 | Regenerative business model transformation framework value creation and delivery.

	Culture and education	Systemic governance and regulatory accountability	Innovation pathways	Network solutions and stakeholder integration
Explanation	We foster a healthy and meaningful workplace culture that aligns with the organization's core values. We create intrinsic motivation to contribute to the organization's goals and broader societal values.	We advocate for systemic governance and regulatory accountability that prioritizes planetary health and societal well-being. By aligning our regenerative values with adaptive policies and collaborative metrics, we foster accountability, resilience, and outcomes that reflect ecological and social interdependence.	We focus on innovation pathways that create regenerative products, services, and processes benefiting both people and the planet. By integrating ecological knowledge, green chemistry, and nature-inspired solutions into our R&D, we align design principles with social–ecological systems interdependence.	We build collaborative networks and integrate diverse stakeholders to address systemic challenges and foster regenerative solutions. By establishing cross-sector partnerships, co-creation platforms, and knowledge-sharing frameworks, we align efforts with ecological interdependence and enhance community well-being.
Transformative questions	What: What workplace initiatives, educational materials, or knowledge-sharing platforms are embedding value creation and delivery around regenerative practices in our day-to-day work? Why: Why does embedding societal and planetary well-being into our core mission lead to more resilient, future-ready business models? Where: Where can we integrate interdisciplinary approaches (e.g., ecology, social sciences, regenerative leadership) to embed nature-inclusive and socially equitable principles into our operations and culture? How: How can we measure and communicate the impact of educational initiatives in reshaping mindsets in advancing equity, ecological restoration, and regenerative value delivery? When: When should we assess and update our value-creation strategies to reflect emerging scientific insights and societal challenges? Who: Who (across communities, partners, policymakers, etc.) should be engaged to ensure holistic perspectives to guide our organizational priorities?	What: What governance frameworks or accountability mechanisms can effectively reflect the value creation and delivery processes while directing them towards enhancing social–ecological systems? Why: Why is embedding ecological interdependence, justice, and fairness into governance structures essential for driving systemic regenerative impact? Where: Where can we implement policy pilots or public-private collaborations to test the embedding of planetary health into governance protocols? How: How can we use transparent reporting, open data, and community-led audits to ensure accountability and reinforce regenerative values? When: When should we align our accountability checkpoints with socio-ecological cycles (e.g., seasonal shifts, economic transitions) to remain relevant and responsive? Who: Who must be represented in governance and oversight structures to ensure our measurement systems reflect ecological and social interdependencies and responsibilities?	What: What innovation design principles can we adopt to create products, services, and supply chains that preserve and enhance social–ecological systems? Why: Why is embedding regeneration into our innovation processes critical when creating and delivering innovation that creates shared value for ecosystems, communities, and society? Where: Where can we create or leverage multi-stakeholder hubs, innovation clusters, or living labs to embed planetary health into collective decision-making? When: When should we iterate product or service development or production to reflect evolving ecological insights and stakeholder feedback? How: How can we leverage green chemistry, biomimicry, and nature-based solutions to create and deliver innovations that regenerate natural systems, and advance societal well-being? Who: Who within our innovation teams and partner networks can guide the development of solutions rooted in social and ecological harmony?	What: What collaborative frameworks or alliances can we develop to share the societal and ecological costs when delivering our products or services? Why: Why do we rely on effective partner and solution networks to create and deliver long-term shared value? Where: Where can we establish nature-inclusive, and co-creative cross-sector partnerships that help us build solution networks enhancing social–ecological systems? Where do we identify suitable partners? How: How can we design transparent, co-creative knowledge-sharing platforms that empower participants to contribute, learn, and drive regenerative action for people, communities, and the planet? When: When and where should we enhance our current sustainability systems to progressively create and deliver more just and fair value to our surroundings and stakeholders? Who: Who among stakeholders can leverage and contribute as a co-creator to build long-lasting and regenerative supply chains when creating and delivering our value proposition?

(Continues)

TABLE 5 | (Continued)

	Culture and education	Systemic governance and regulatory accountability	Innovation pathways	Network solutions and stakeholder integration
Example	Phipps Conservatory and Botanical Gardens built the Center for Sustainable Landscapes, a net-positive energy and net-zero water building that provides a healthy and inspiring work environment (Phipps Conservatory and Botanical Gardens 2024).	Interface, a global carpet tile manufacturer, has established processes to manage time, resources, and waste through its “Mission Zero” initiative, aimed at eliminating its environmental impact by 2020. They optimize natural capital regeneration by using recycled materials and sustainable practices, such as closed-loop manufacturing (Fedrizzi et al. 2020).	Notpla creates biodegradable food packaging from seaweed, addressing environmental challenges by providing an alternative to plastic and supporting seaweed farming communities (Doan 2023).	Patagonia’s founder Yvon Chouinard declared, “Earth is now our only shareholder”, restructuring the company to dedicate profits to ecosystem conservation and regeneration (Patagonia 2021).

3) emphasizes developing RBM strategies that overcome market and financial barriers and translate regenerative principles into operational practice. *Networked Solutions and Stakeholder Integration* (Dimension 4) highlights the need for cross-sector collaboration and alignment among industry actors, academia, communities, and policymakers to drive systemic change. Together, these four dimensions provide a comprehensive framework for guiding firms through the process of transforming existing business models into regenerative ones, emphasizing both organizational and systemic renewal.

4.7 | Evaluation and Iteration of Transformation Outcomes

As a third and final step, firms evaluate the implementation of each initiative that is generated throughout the transformation process. To assess the improvements and the progress made, the following questions guide the evaluation: “How have the prioritized areas improved? Where does the next improvement need to happen? What consequences does one improvement have on another, reflecting the interconnectedness of regenerative systems?” (see Figure 3).

4.8 | Demonstration

To evaluate our proposed solution, we demonstrated the process template in a workshop setting (Dimov et al. 2023; Seckler et al. 2021; Venable et al. 2016). In doing so, we were able to assess the usefulness of the solution by collecting feedback from seven workshop participants. We adopted a “Quick and Simple” evaluation strategy as outlined in the *FEDS Framework for Evaluation in Design Science Research* (Venable et al. 2016). Based on this framework, we evaluated the artifact’s practical usefulness in three ways: formatively, artificially, and naturalistically. Figure 4 shows the initial version of our process template prior to the demonstration workshop, which underwent significant refinement based on participant feedback.

Following Venable et al. (2016), a formative evaluation involves improving the artifact under evaluation to achieve a better outcome. We refined the process template and complementary questionnaire based on participant feedback. An artificial evaluation involves conceptual or rhetorical testing rather than empirical hypothesis verification. Here, we presented and discussed the artifact’s logic and usability to identify conceptual and practical gaps. Finally, a naturalistic evaluation assesses the artifact in a real-world context. In our case, this involved reflecting on how the developed tool could be applied within organizational environments. To illustrate the improvements resulting from the demonstration stage, we present both the initial iteration (Figure 4) and the final enhanced version (Figure 3) of our process template, which incorporates all post-demonstration refinements.

Subsequently, to refine the process template and the complementary questionnaire, we engaged in a series of discussions on how to iteratively improve our solution. These sessions focused

TABLE 6 | Regenerative business model transformation framework value capture.

	Culture and education	Systemic governance and regulatory accountability	Innovation pathways	Network solutions and stakeholder integration
Explanation	We foster a healthy and meaningful workplace culture that aligns with the organization's core values. We create intrinsic motivation to contribute to the organization's goals and broader societal values.	We advocate for systemic governance and regulatory accountability that prioritizes planetary health and societal well-being. By aligning our regenerative values with adaptive policies and collaborative metrics, we foster accountability, resilience, and outcomes that reflect ecological and social interdependence.	We focus on innovation pathways that create regenerative products, services, and processes benefiting both people and the planet. By integrating ecological knowledge, green chemistry, and nature-inspired solutions into our R&D, we align design principles with social-ecological systems interdependence.	We build collaborative networks and integrate diverse stakeholders to address systemic challenges and foster regenerative solutions. By establishing cross-sector partnerships, co-creation platforms, and knowledge-sharing frameworks, we align efforts with ecological interdependence and enhance community well-being.
Transformative questions	What: What organizational culture do we envision that centers around regeneration and what do we need to teach so that individuals strive for net-positive ecological and social outcomes? Why: Why is aiming for a net-positive impact essential to move beyond sustainability as a compliance exercise and move collectively toward true planetary stewardship? Where: Where can we integrate examples of net-positive initiatives (e.g., regenerative agriculture, circular production models, green chemistry best practices) into existing educational frameworks? How: How can we embed regenerative metrics and success stories in our day-to-day operations and our educational content to inspire others to set and achieve net-positive goals? When: When should we evaluate the effectiveness of mindset shifts to ensure that learning leads to genuine regenerative action over time? Who: Who in our training and development teams can champion a shift from preserve approaches to actively regenerative 'enhancing' thinking?	What: What multiple accounting frameworks or standards can be adapted or created to measure progress toward net-positive impact? Why: Why does striving for net-positive impact require more ambitious and forward-looking accountability frameworks than traditional sustainability targets? Why do we need to involve policymakers in this process? Where: Where should we invest in monitoring technology or partner with local stakeholders to accurately measure regenerative outcomes? How: How can we incorporate regenerative principles into our revision and compliance strategies, ensuring that multiple accounting mechanisms guide continuous improvement? When: When should we adjust our governance models to reflect lessons learned on achieving net-positive results? Who: Who needs to be involved in defining net-positive targets, and who will ensure adherence to these goals over time?	What: What financial models, incentive structures, or timeframes are needed to scale net-positive innovations despite market pressures? Why: Why is targeting net-positive impact future-proofing our business model? Where: Where can we integrate regenerative ingredients, materials, or processes to ensure that each innovation contributes to overall regeneration? How: How can we continuously refine and iterate innovations so that they actively restore ecosystems and uplift communities? When: When should we launch, iterate, or retire innovations to maintain a trajectory toward net-positive outcomes? Who: Who can champion transformative ideas that move beyond incremental improvements toward net-positive technologies and strategies?	What: What common principles, guidelines, or commitments can we adopt to ensure that our collective action produces more regenerative outcomes than any single actor could achieve? Why: Why is a networked, integrated approach essential to achieve a net-positive impact? Where: Where can we pool resources, knowledge, and influence to scale up net-positive innovations across regions, sectors, and value chains? How: How can we ensure that each collaborative initiative makes a tangible contribution to regenerating ecosystems, empowering communities, and shifting cultural narratives? When: When should we coordinate efforts to align with natural cycles, policy windows, or market shifts that favor regenerative transitions? Who: Who among our partners and collaborators can help us set ambitious targets for net-positive outcomes and hold each other accountable?

(Continues)

TABLE 6 | (Continued)

Culture and education		Systemic governance and regulatory accountability		Innovation pathways	Network solutions and stakeholder integration
Example	Phipps Conservatory and Botanical Gardens built the Center for Sustainable Landscapes, a net-positive energy and net-zero water building that provides a healthy and inspiring work environment (Phipps Conservatory and Botanical Gardens 2024).	Interface, a global carpet tile manufacturer, has established processes to manage time, resources, and waste through its “Mission Zero” initiative, aimed at eliminating its environmental impact by 2020. They optimize natural capital regeneration by using recycled materials and sustainable practices, such as closed-loop manufacturing (Fedrizzi et al. 2020).		Notpla creates biodegradable food packaging from seaweed, addressing environmental challenges by providing an alternative to plastic and supporting seaweed farming communities (Doan 2023).	Patagonia’s founder Yvon Chouinard declared, “Earth is now our only shareholder”, restructuring the company to dedicate profits to ecosystem conservation and regeneration (Patagonia 2021).

on demonstrating the solution to potential users, clarifying key components, and responding to unclear aspects. To illustrate a post-transformation RBM, we added specific examples that demonstrate how successfully transformed parts of a business model can look like. We therefore resorted to the identified regenerative business cases by Das and Bocken (2024) and further web-based research (which can be found in Tables 4, 5, 6 as examples). The participant feedback was collected via notes, allowing for collaborative adjustment of specific sections of the process template. Insights from this workshop were then used to justify the necessary adjustments and guide further development of the solution (see Tables 4, 5, 6). This cycle resulted in the development of the final and adapted version of the RBM transformation process template and the transformative questionnaire.

The final version of the artifact integrates several key improvements. First, we improved the usability of the process template by simplifying the description of the individual process steps and further clarifying the self-assessment section. This helps users to identify and pursue suitable transformation pathways. Second, we embedded a diagnostic questionnaire that is designed to prompt critical reflection on areas requiring regenerative change. These questions now serve as a core component of the process template. Third, we integrated several practical examples based on the work of Das and Bocken (2024) and further web-based research to illustrate what an RBM can look like in practice. This inclusion offers users concrete reference points and best-practice cases, which makes the concept of regenerative transformation more accessible. The results indicate that the solution represents a useful tool for transforming non-regenerative to RBMs (as shown in Figure 3).

5 | Discussion

This study aimed to explore how incumbents in RIIs can move beyond incremental sustainability-oriented change and transform their current business models toward regeneration. In doing so, we examined the mechanism that enables firms to transform from incremental sustainability improvements to RBMs that unlock systemic change. To address this, we employed a DSRM (Dimov et al. 2023; Peffers et al. 2007; Tuunanen et al. 2024) and developed a three-step process template that assesses a firm’s readiness for an RBM transformation, designs and implements corresponding interventions, and evaluates the progress achieved. This artifact is further complemented by a diagnostic and targeted questionnaire that supports the design and implementation of an RBM transformation initiative.

These artifacts serve as practically useful solutions to the problem we have identified, such as the mainly linear and open-ended business models of incumbents in RIIs, which cause major environmental and social harm. Thus, our study has important practical implications that equip managers with the relevant tools to overcome the inherent sustainability challenges of RIIs. Further, this study makes equally important methodological and theoretical contributions to the literature on RBM innovation and the broader sustainability-oriented management literature. More specifically, this study responds to calls for further research on the practical adoption

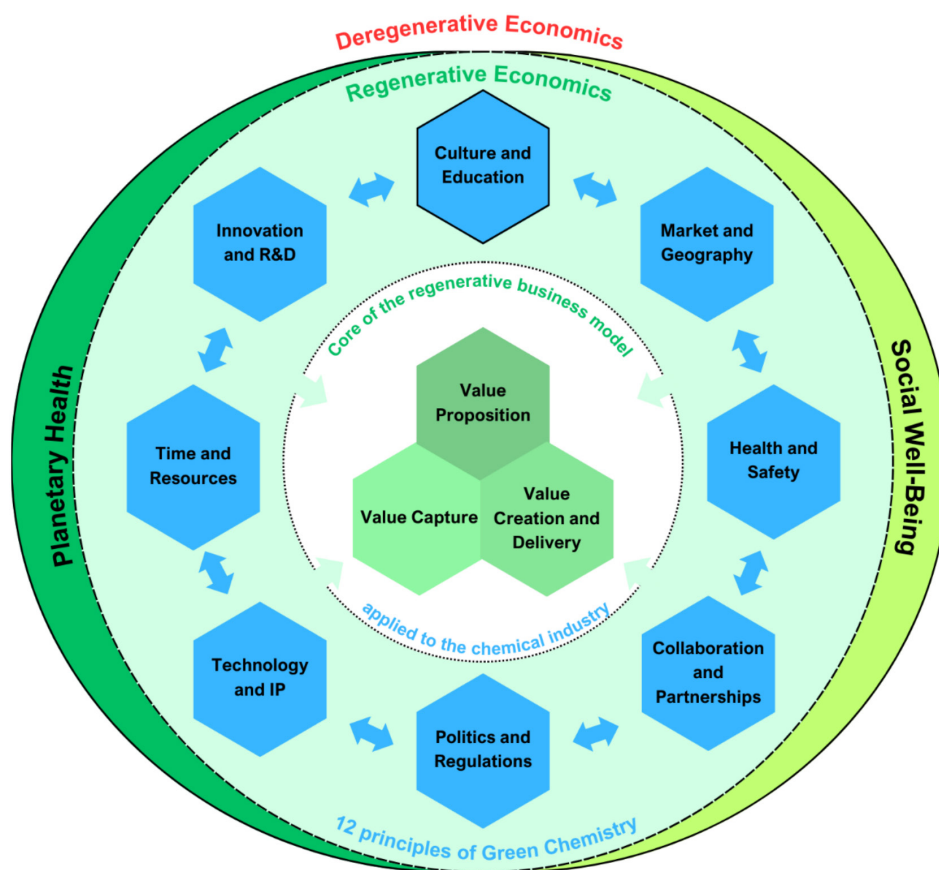


FIGURE 4 | First iteration before demonstration of “regenerative business model tool”.

of RBMs (Das and Bocken 2024; Hahn and Tampe 2021; Konietzko et al. 2023).

5.1 | Theoretical Contributions

Our study makes several theoretical contributions to the literature on RBMs (e.g., Konietzko et al. 2023) and systemic business model transformation (e.g., Frishammar and Parida 2019). While regeneration and RBMs have emerged as concepts to rethink current sustainability ideas, they lack sufficient clarification to enable a transformation toward regeneration. Thus, we expand the literature on RBMs by highlighting the essential organizational mechanisms that facilitate an RBM transformation (Das and Bocken 2024; Konietzko et al. 2023). Our study theorizes a dynamic transformation process in which firms need to re-ground, re-wire, and re-allocate their existing business models to transition to RBMs. To re-ground, a firm needs to transform its current value proposition by moving from a perception of harm reduction toward a commitment to the systemic enhancement of social–ecological systems. Second, firms need to re-wire their organizational activity to co-create and co-deliver value in a just and fair way to its community and regional ecosystems. Third, they need to re-seize value through metrics and governance mechanisms that tie economic returns to regenerative multi-capital accounting. This enables firms to reconfigure their value proposition, value creation and delivery, as well as their value capture to transform business model practices aimed at delivering net-positive outcomes (Konietzko et al. 2023). In

doing so, we identify the organizational mechanisms and regenerative capabilities that enable incumbents to move from incremental improvements to systemic regenerative change. This complements previous studies in the sustainability and circular economy discourse (e.g., Frishammar and Parida 2019; Müller and Remke 2025; Trunk et al. 2025).

While previous studies on *business sustainability transformation* have emphasized the importance of shifting organizational narratives toward the “common good” and broadening the scope of environmental impacts beyond the organizational boundaries (e.g., Hermelingmeier and Von Wirth 2021), they offer limited explanations of how this broader perspective can be integrated into organizational boundaries that extend into social–ecological systems. Moreover, the embedding and enactment of a holistic organizational perspective within the business model remains underexplored. This has implications for understanding the mechanisms that enable such transformative change. We address these shortcomings by clarifying and providing the mechanisms required to unlock and guide systemic business model transformations for regeneration (e.g., Hausdorf and Timm 2023; Hermelingmeier and Von Wirth 2021; Konietzko et al. 2023). By articulating a multi-level process, our study provides the missing clarification that unlocks the potential of regeneration. In doing so, we identify the organizational mechanisms and regenerative capabilities that enable incumbents to move from incremental improvements to systemic regenerative change. This complements previous studies on driving sustainability for incumbents (e.g.,

Bohnsack et al. 2020; Frishammar and Parida 2019; Wesseling et al. 2015).

In addition, our work directly responds to the call by Zimmerman et al. (2020) for an integrated systemic framework to foster a green chemistry future. At the same time, we expand our findings from the research context of the chemical industry to the broader field of RIIs. While existing sustainability-oriented transformation efforts in the chemical industry emphasize incremental process improvements, we advocate for a holistic, inclusive, and co-creative transformation approach across all RIIs. This shift acknowledges that organizational change is not isolated but deeply embedded in and influential upon broader social–ecological systems to generate a large-scale systemic transformation (Kulp and Mrożewski 2025; Matlin et al. 2016; Zimmerman et al. 2020). By integrating a systems perspective (Buckton et al. 2023), our process template addresses these systemic interdependencies and provides a structured yet flexible approach to enable regenerative practices that extend beyond the boundaries of individual firms. This system level orientation is crucial in tackling the complex, multidimensional challenges encountered by RIIs (Ghosh and Bakshi 2017).

5.2 | Methodological Contribution

We offer several methodological contributions that advance the application of DSRM to the domain of business model transformation and sustainability research, especially for the RBM literature. First, we demonstrate how DSRM can be adapted for the specific context of business model transformation in RIIs. While previous design science studies in sustainability research have focused primarily on product or process innovation, we show how the six-step DSRM process (Peffer et al. 2007) can be tailored to address the systemic, multilevel challenges of transforming entire business models toward regeneration. Second, we contribute to methodological rigor in design science research by integrating multiple data sources and evaluation methods. Our approach combines semi-structured interviews with three distinct stakeholder groups, supplementary organizational documents, and a workshop-based formative evaluation (Venable et al. 2016). This triangulation strengthens the validity of our DRs and enhances confidence in the utility of our artifacts. We demonstrate how qualitative data analysis techniques, specifically the Gioia methodology (Gioia et al. 2013) and a thematic analysis (Braun and Clarke 2006), can be systematically integrated into the DSRM process to derive DRs from empirical data. Finally, we develop and apply specific evaluation criteria for RBM artifacts. Building on Van Aken and Berends' (2018) framework for organizational problem-solving, we establish a structured approach for translating qualitative insights into actionable DRs across four categories: functional requirements, user requirements, boundary conditions, and design restrictions. This provides a replicable template for future DSRM in regeneration and sustainability contexts.

5.3 | Practical Implications

Our study offers important practical implications. We provide practitioners with a useful three-step process template that

enables a sustainability-focused business model transformation by assessing the RBM readiness, designing interventions, and evaluating progress. This co-creative tool allows for the integration of a holistic social–ecological systems perspective into RIIs (Matlin et al. 2016; Zimmerman et al. 2020). Further, we demonstrated the artifact to practitioners. As a consequence of iterative rounds of demonstration, our regenerative transformation tool was deemed suitable and practically useful to initiate an RBM transformation (Peffer et al. 2007).

At the same time, our solution addresses the practical challenges of RIIs, particularly the chemical industry, where long innovation cycles, resistance to change, and strong dependence on policy directives pose barriers to sustainability-oriented innovation. While incremental progress toward sustainability has been achieved, our findings show that managers must recognize the interconnectedness of their actions within social–ecological systems to advance regeneration in a meaningful way. The process model we developed supports practitioners in understanding these interdependencies and provides concrete guidance for designing and implementing interventions for transformative, systemic, and regenerative change.

Moreover, organizations implementing our process template should anticipate specific resource commitments and timelines to ensure successful transformation. The initial readiness phase (Step 1) typically requires 2–3 months and involves forming a cross-functional team of five to eight people with executive sponsorship to conduct stakeholder interviews and evaluate the organization's current position along the degenerative-to-regenerative spectrum. The intervention phase (Step 2) demands 4–6 months of dedicated effort, including budget allocation for pilot projects and potential partnerships with external regeneration experts or research institutions to test regenerative initiatives in controlled settings before scaling. Organizations should expect the evaluation phase (Step 3) to be ongoing, requiring continuous monitoring systems and quarterly review cycles to track progress and adapt strategies.

A crucial success factor we identified is addressing resistance from traditional mindsets through systematic education programs using our tool's Culture and Education dimension. This dimension starts with small pilot projects in one business unit and helps to build organizational confidence through demonstrated results rather than attempting wholesale transformation immediately. Moreover, firms should view regenerative transformation as a business opportunity rather than just a cost, by identifying initiatives where regenerative outcomes align with financial performance. By following this phased approach with realistic resource allocation, incumbents in RIIs can move beyond reactive sustainability compliance toward proactive regenerative transformation that actively benefits the social–ecological systems in which they operate.

In addition to the implications for practitioners using the process model to transform their business models, we also offer recommendations for policymakers. Our analysis revealed that key barriers to transformation stem from a fixed mindset toward further sustainability-oriented change and from the absence of generalized metrics that ensure comparative

transparency. While interviewees acknowledged that regulation such as sustainability standards have improved the overall quality of transformation efforts, many emphasized that these standards are critical drivers of sustainability across the industry. Therefore, we recommend that policymakers develop a common standard for multi-capital accounting that captures not only financial and environmental dimensions but also integrates societal metrics to promote communal well-being and social progress. Finally, given the long innovation cycles within the chemical industry, we propose advancing toward regenerative standards that encourage firms to rethink their current sustainability practices and incentivize early adopters to lead this transition.

In summary, this study proposes a practically useful tool that aids firms in transforming their business models toward regeneration. In doing so, we support firms in aligning their business model practices with social–ecological systems and in actively advancing their development. By enabling the widespread adoption of regenerative practices through a corresponding business model structure, our tool helps firms take important steps toward a more sustainable future.

5.4 | Limitations and Directions for Further Research

We acknowledge that this study has several limitations. At the same time, this opens up avenues for future research. First, we collected data from three expert groups. This limits the depth of insight within each group, yet it aligns with our research approach of gaining a holistic and multi-stakeholder understanding. This approach enabled us to collect a rich dataset that captures different perspectives along the chemical industry's value chain. Yet it creates opportunities for future research. Future studies could expand the interviews within each stakeholder category and conduct additional demonstration workshops. In this way, future research could generate richer feedback for refining and quantitatively testing the developed solution for broader generalization. This could enable the development of more customized process models tailored to specific contexts.

Second, we focused specifically on the chemical industry as a representative of RIIs, thus our findings remain bound to this research context. While several characteristics of the chemical industry can be observed across other RIIs, the applicability of our artifacts may be limited. Future research could therefore engage in studying application scenarios beyond the chemical industry to assess the suitability of our proposed solution across different sectors.

Third, the chosen methodology limits the depth of exploration in the examined context (Peffer et al. 2007; Tuunanen et al. 2024). For example, using an infinity methodology (Seckler et al. 2023) could allow for a deeper investigation into various sectors, including small firms and new ventures.

Fourth, RBMs are still in the early stages of theoretical and practical development. In this study, we contributed to their

enactment by re-grounding, re-wiring, and re-seizing business models for regeneration. However, there remain several future pathways for research and practice to deepen knowledge on this topic. Future research should build and expand on the various conceptualizations of regeneration that appear in specific operational practices and performance metrics. While the conceptual foundations of RBMs have been established (Hahn and Tampe 2021; Konietzko et al. 2023), moving beyond mere definitional and conceptual clarity toward detailed operationalization remains an important frontier. We propose that future research should examine areas such as: first, how do regenerative principles apply in context-specific practices across various industries and regions. This line of inquiry can extend to different sectors and to various stages of venture development. Second, although multi-capital accounting has been suggested as a foundation for regenerative assessment and value capture, future research should investigate which operational metrics effectively capture net-positive impacts on social–ecological systems. This assessment requires careful consideration of quantitative indicators together with qualitative dimensions that reflect the enhancement of social–ecological systems. Lastly, we encourage future scholarship to assess how regenerative outcomes can be evaluated and compared across different organizational settings. Practices viewed as regenerative in one context may be assessed differently in another, which highlights the need to define clearer criteria for the comparability of regenerative outcomes and thresholds. This implementation-focused research agenda builds on the conceptual base already established in the literature while addressing the practical knowledge gaps that currently hinder widespread adoption.

6 | Conclusion

In this paper, we address how incumbent firms in RIIs can transform their business models for regeneration. While scholarship on RBMs has advanced theoretical understanding, firms still struggle to translate these ideas into practice and to develop the transformative capabilities required for implementation. We extend this line of inquiry by adopting a design science approach, shifting the focus from defining *what* RBMs are to understanding *how* existing business models can be transformed toward regeneration. We develop and propose a three-step process template and a complementary questionnaire that guides firms through an RBM transformation. We demonstrate that regeneration and resource intensity, though seemingly incompatible, can be deeply interconnected when approached systemically. With our research, we contribute to the literature on RBMs by bridging the conceptual understanding and practical application. In this way, we theorize a dynamic process in which firms need to *re-ground*, *re-wire*, and *re-seize* their business models to achieve regenerative transformation. Therefore, we demonstrate that the transformation of business models to foster regeneration necessitates a paradigmatic shift in mindset, involving the unlearning of traditional business principles, and reconsideration of social–ecological relationships. Nevertheless, such transformation is attainable through systemic thinking and organizational commitment.

Author Contributions

Jan-Friedrich Kulp: conceptualization, methodology, formal analysis, investigation, writing – original draft, writing – review and editing, visualization. **Konstantin Remke:** methodology, formal analysis, investigation, writing – original draft, writing – review and editing. **Yacine Cherraoui:** conceptualization, methodology, data collection, formal analysis, investigation, writing – original draft. **Jill Richard Kickul:** conceptualization, formal analysis, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

References

- Andersson, F. N. G. 2020. "Effects on the Manufacturing, Utility and Construction Industries of Decarbonization of the Energy-Intensive and Natural Resource-Based Industries." *Sustainable Production and Consumption* 21: 1–13. <https://doi.org/10.1016/j.spc.2019.10.003>.
- Bag, S., S. Gupta, C. Galera-Zarco, and I. Laguir. 2025. "Developing Green Hydrogen-Based Regenerative Supply Chain Strategies for a Sustainable Future: An Empirical Study." *Business Strategy and the Environment* 34: 6968–6990. <https://doi.org/10.1002/bse.4334>.
- Bag, S., and M. S. Rahman. 2023. "Navigating Circular Economy: Unleashing the Potential of Political and Supply Chain Analytics Skills Among Top Supply Chain Executives for Environmental Orientation, Regenerative Supply Chain Practices, and Supply Chain Viability." *Business Strategy and the Environment* 33, no. 2: 504–528. <https://doi.org/10.1002/bse.3507>.
- Baldassarre, B., J. Konietzko, P. Brown, et al. 2020. "Addressing the Design-Implementation Gap of Sustainable Business Models by Prototyping: A Tool for Planning and Executing Small-Scale Pilots." *Journal of Cleaner Production* 255: 120295. <https://doi.org/10.1016/j.jclepro.2020.120295>.
- Bansal, P., Tima, R. Durand, M. Kreutzer, S. Kunisch, and A. M. McGahan. 2025. "Strategy Can No Longer Ignore Planetary Boundaries: A Call for Tackling Strategy's Ecological Fallacy." *Journal of Management Studies* 62, no. 2: 965–985. <https://doi.org/10.1111/joms.13088>.
- Bellato, L., N. Frantzeskaki, and C. A. Nygaard. 2023. "Regenerative Tourism: A Conceptual Framework Leveraging Theory and Practice." *Tourism Geographies* 25, no. 4: 1026–1046. <https://doi.org/10.1080/14616688.2022.2044376>.
- Birbrair, A., T. Zhang, Z.-M. Wang, et al. 2013. "Role of Pericytes in Skeletal Muscle Regeneration and Fat Accumulation." *Stem Cells and Development* 22, no. 16: 2298–2314. <https://doi.org/10.1089/scd.2012.0647>.
- Bocken, N. M. P., S. W. Short, P. Rana, and S. Evans. 2014. "A Literature and Practice Review to Develop Sustainable Business Model Archetypes." *Journal of Cleaner Production* 65: 42–56. <https://doi.org/10.1016/j.jclepro.2013.11.039>.
- Bohnsack, R., A. Kolk, J. Pinkse, and C. M. Bidmon. 2020. "Driving the Electric Bandwagon: The Dynamics of Incumbents' Sustainable Innovation." *Business Strategy and the Environment* 29, no. 2: 727–743. <https://doi.org/10.1002/bse.2430>.
- Boons, F., and F. Lüdeke-Freund. 2013. "Business Models for Sustainable Innovation: State-of-the-Art and Steps Towards a Research Agenda." *Journal of Cleaner Production* 45: 9–19. <https://doi.org/10.1016/j.jclepro.2012.07.007>.
- Boons, F., C. Montalvo, J. Quist, and M. Wagner. 2013. "Sustainable Innovation, Business Models and Economic Performance: An Overview." *Journal of Cleaner Production* 45: 1–8. <https://doi.org/10.1016/j.jclepro.2012.08.013>.
- Branzei, O., P. Munoz, S. Russell, and G. Whiteman. 2017. "Call for Papers: Special Issue on "Regenerative Organizations: Business and Climate Action Beyond Mitigation and Adaptation"." *Organization & Environment* 30, no. 3: 275–277. <https://doi.org/10.1177/108602661728990>.
- Braun, V., and V. Clarke. 2006. "Using Thematic Analysis in Psychology." *Qualitative Research in Psychology* 3, no. 2: 77–101. <https://doi.org/10.1191/1478088706qp0630a>.
- Buckton, S. J., I. Fazey, B. Sharpe, et al. 2023. "The Regenerative Lens: A Conceptual Framework for Regenerative Social-Ecological Systems." *One Earth* 6, no. 7: 824–842. <https://doi.org/10.1016/j.oneear.2023.06.006>.
- Caldera, S., S. Hayes, L. Dawes, and C. Desha. 2022. "Moving Beyond Business as Usual Toward Regenerative Business Practice in Small and Medium-Sized Enterprises." *Frontiers in Sustainability* 3: 799359. <https://doi.org/10.3389/frsus.2022.799359>.
- Carter, N., D. Bryant-Lukosius, A. DiCenso, J. Blythe, and A. J. Neville. 2014. "The Use of Triangulation in Qualitative Research." *Oncology Nursing Forum* 41, no. 5: 545–547. <https://doi.org/10.1188/14.ONF.545-547>.
- Center for Sustainable Landscapes|Phipps Conservatory and Botanical Gardens|Pittsburgh PA. 2024. "Phipps Conservatory." <https://www.phippsconservatory.org/green-innovation/at-hipps/center-for-sustainable-landscapes-greenest-building-museum-garden-in-the-world>.
- Civera, C., G. Santoro, and V. Chiaudano. 2025. "Do Not Stop the Loop: Driving Circular Textiles With Technology and Stakeholder Engagement." *Business Strategy and the Environment* 34, no. 5: 5317–5335. <https://doi.org/10.1002/bse.4250>.
- Das, A., and N. Bocken. 2024. "Regenerative Business Strategies: A Database and Typology to Inspire Business Experimentation Toward Sustainability." *Sustainable Production and Consumption* 49: 529–544. <https://doi.org/10.1016/j.spc.2024.06.024>.
- De Faria, D. R. G., J. L. De Medeiros, and O. Q. F. Araújo. 2021. "Sustainability Assessment for the Chemical Industry: Onwards to Integrated System Analysis." *Journal of Cleaner Production* 278: 123966. <https://doi.org/10.1016/j.jclepro.2020.123966>.
- De Magalhães, C. 2015. In *Urban Regeneration, International Encyclopedia of the Social & Behavioral Sciences*, 919–925. Elsevier. <https://doi.org/10.1016/b978-0-08-097086-8.74031-1>.
- Dean, T. J., and J. S. McMullen. 2007. "Toward a Theory of Sustainable Entrepreneurship: Reducing Environmental Degradation Through Entrepreneurial Action." *Journal of Business Venturing* 22, no. 1: 50–76. <https://doi.org/10.1016/j.jbusvent.2005.09.003>.
- Dietze, M. C., and J. S. Clark. 2008. "Changing the Gap Dynamics Paradigm: Vegetative Regeneration Control on Forest Response to Disturbance." *Ecological Monographs* 78, no. 3: 331–347. <https://doi.org/10.1890/07-0271.1>.
- Dimov, D., M. Maula, and A. G. L. Romme. 2023. "Crafting and Assessing Design Science Research for Entrepreneurship." *Entrepreneurship*

- Theory and Practice 47, no. 5: 1543–1567. <https://doi.org/10.1177/10422587221128271>.
- Doan, H. 2023. "Impact Report 2023.pdf. Google Docs - Noptla Impact Report." <https://drive.google.com/file/d/1aYqZZ13VxPVC-tKufD8S30MU5obFv5pY/view>.
- Dredge, D. 2022. "Regenerative Tourism: Transforming Mindsets, Systems and Practices." *Journal of Tourism Futures* 8, no. 3: 269–281. <https://doi.org/10.1108/jtf-01-2022-0015>.
- Du Plessis, C. 2012. "Towards A Regenerative Paradigm for the Built Environment." *Building Research and Information* 40, no. 1: 7–22. <https://doi.org/10.1080/09613218.2012.628548>.
- Eisenhardt, K. M., M. E. Graebner, and S. Sonenshein. 2016. "Grand Challenges and Inductive Methods: Rigor Without Rigor Mortis." *Academy of Management Journal* 59, no. 4: 1113–1123. <https://doi.org/10.5465/amj.2016.4004>.
- Etikan, I. 2016. "Comparison of Convenience Sampling and Purposive Sampling." *American Journal of Theoretical and Applied Statistics* 5, no. 1: 1. <https://doi.org/10.11648/j.ajtas.20160501.11>.
- Fath, B. D., D. A. Fiscus, S. J. Goerner, A. Berea, and R. E. Ulanowicz. 2019. "Measuring Regenerative Economics: 10 Principles and Measures Undergirding Systemic Economic Health." *Global Transitions* 1: 15–27. <https://doi.org/10.1016/j.glt.2019.02.002>.
- Fayed, A. E., and R. Rashed. 2020. *Aspects of Regenerative Cities, Architecture and Urbanism: A Smart Outlook*, 303–319. Springer International Publishing. https://doi.org/10.1007/978-3-030-52584-2_22.
- Fedrizzi, R., R. Anderson, and J. Makower. 2020. "Lessons for the Future." In *Mission Zero*. https://www.interface.com/content/dam/interfaceinc/interface/sustainability/emea/25th-anniversary-report/Interface_MissionZeroCel_Booklet_EN.pdf.
- Folke, C., S. R. Carpenter, B. Walker, M. Scheffer, T. Chapin, and J. Rockström. 2010. "Resilience Thinking: Integrating Resilience, Adaptability and Transformability." *Ecology and Society* 15, no. 4: 20.
- Frishammar, J., and V. Parida. 2019. "Circular Business Model Transformation: A Roadmap for Incumbent Firms." *California Management Review* 61, no. 2: 5–29. <https://doi.org/10.1177/0008125618811926>.
- Geissdoerfer, M., M. P. P. Pieroni, D. C. A. Pigosso, and K. Soufani. 2020. "Circular Business Models: A Review." *Journal of Cleaner Production* 277: 123741. <https://doi.org/10.1016/j.jclepro.2020.123741>.
- Geissdoerfer, M., T. Santa-Maria, J. Kirchherr, and C. Pelzeter. 2023. "Drivers and Barriers for Circular Business Model Innovation." *Business Strategy and the Environment* 32, no. 6: 3814–3832. <https://doi.org/10.1002/bse.3339>.
- Geissdoerfer, M., P. Savaget, N. M. P. Bocken, and E. J. Hultink. 2017. "The Circular Economy—A New Sustainability Paradigm?" *Journal of Cleaner Production* 143: 757–768. <https://doi.org/10.1016/j.jclepro.2016.12.048>.
- Ghosh, T., and B. R. Bakshi. 2017. "Process to Planet Approach to Sustainable Process Design: Multiple Objectives and Byproducts." *Theoretical Foundations of Chemical Engineering* 51, no. 6: 936–948. <https://doi.org/10.1134/s0040579517060045>.
- Gioia, D. A., K. G. Corley, and A. L. Hamilton. 2013. "Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology." *Organizational Research Methods* 16, no. 1: 15–31. <https://doi.org/10.1177/1094428112452151>.
- Gonzalez Hernandez, A., L. Paoli, and J. M. Cullen. 2018. "How Resource-Efficient is the Global Steel Industry?" *Resources, Conservation and Recycling* 133: 132–145. <https://doi.org/10.1016/j.resconrec.2018.02.008>.
- Gregor, S., and A. R. Hevner. 2013. "Positioning and Presenting Design Science Research for Maximum Impact." *MIS Quarterly* 37, no. 2: 337–355. <https://doi.org/10.25300/misq/2013/37.2.01>.
- Gualandris, J., O. Branzei, M. Wilhelm, et al. 2024. "Unchaining Supply Chains: Transformative Leaps Toward Regenerating Social–Ecological Systems." *Journal of Supply Chain Management* 60, no. 1: 53–67. <https://doi.org/10.1111/jscm.12314>.
- Hahn, T., and M. Tampe. 2021. "Strategies for Regenerative Business." *Strategic Organization* 19, no. 3: 456–477. <https://doi.org/10.1177/1476127020979228>.
- Hart, S. L., and M. B. Milstein. 1999. "Global Sustainability and the Creative Destruction of Industries." *MIT Sloan Management Review* 41, no. 1: 23–33.
- Hausdorf, M., and J. Timm. 2023. "Business Research for Sustainable Development: How Does Sustainable Business Model Research Reflect Doughnut Economics?" *Business Strategy and the Environment* 32, no. 6: 3398–3416. <https://doi.org/10.1002/bse.3307>.
- Hermelingmeier, V., and T. Von Wirth. 2021. "The Nexus of Business Sustainability and Organizational Learning: A Systematic Literature Review to Identify Key Learning Principles for Business Transformation." *Business Strategy and the Environment* 30, no. 4: 1839–1851. <https://doi.org/10.1002/bse.2719>.
- Hernández, M., and P. Muñoz. 2022. "Reformists, Decouplists, and Activists: A Typology of Ecocentric Management." *Organization & Environment* 35, no. 2: 282–306. <https://doi.org/10.1177/1086026621993204>.
- Hevner, A. R., S. T. March, J. Park, and S. Ram. 2004. "Design Science in Information Systems Research." *MIS Quarterly* 28, no. 1: 75–106. <https://doi.org/10.2307/25148625>.
- Hutchins, G., and L. Storm. 2019. *Regenerative Leadership: The DNA of Life-Affirming 21st Century Organizations*. Wordzworth Publishing.
- Jovanović-Vujatović, M., L. Bonić, and B. Krstić. 2024. "Key Aspects of Regenerative Business Model: Concept, Principles and Strategies." *Journal of Regenerative Economics* 1, no. 2: 161–175. <https://doi.org/10.5937/jre2402161J>.
- Joyce, A., and R. L. Paquin. 2016. "The Triple Layered Business Model Canvas: A Tool to Design More Sustainable Business Models." *Journal of Cleaner Production* 135: 1474–1486. <https://doi.org/10.1016/j.jclepro.2016.06.067>.
- Kirchherr, J., N.-H. N. Yang, F. Schulze-Spüntrup, M. J. Heerink, and K. Hartley. 2023. "Conceptualizing the Circular Economy (Revisited): An Analysis of 221 Definitions." *Resources, Conservation and Recycling* 194: 107001. <https://doi.org/10.1016/j.resconrec.2023.107001>.
- Konietzko, J., A. Das, and N. Bocken. 2023. "Towards Regenerative Business Models: A Necessary Shift?" *Sustainable Production and Consumption* 38: 372–388. <https://doi.org/10.1016/j.spc.2023.04.014>.
- Kulp, J.-F., and M. Mroźewski. 2025. "The Role of Institutions and Digital Infrastructure for Sustainable Entrepreneurship: A Macro-Level Perspective on Twin Transformation." *International Journal of Innovation Management* 29, no. 05n06: 2540005. <https://doi.org/10.1142/S1363919625400055>.
- Laurencin, C. T., and Y. Khan. 2012. "Regenerative Engineering." *Science Translational Medicine* 4, no. 160: 160ed9. <https://doi.org/10.1126/scitranslmed.3004467>.
- Lewin, K. 1947. *Field Theory in Social Science*. Harper & Row.
- Liu, J., H. Mooney, V. Hull, et al. 2015. "Systems Integration for Global Sustainability." *Science* 347, no. 6225: 1258832. <https://doi.org/10.1126/science.1258832>.
- Lüdeke-Freund, F. 2010. "Towards a Conceptual Framework of 'Business Models for Sustainability'." <https://doi.org/10.13140/RG.2.1.2565.0324>.
- Martinez-Hernandez, E. 2017. "Trends in Sustainable Process Design—From Molecular to Global Scales." *Current Opinion in Chemical Engineering* 17: 35–41. <https://doi.org/10.1016/j.coche.2017.05.005>.

- Mason, C., and P. Dunnill. 2008. "A Brief Definition of Regenerative Medicine." *Regenerative Medicine* 3, no. 1: 1–5. <https://doi.org/10.2217/17460751.3.1.1>.
- Matlin, S. A., G. Mehta, H. Hopf, and A. Krief. 2016. "One-World Chemistry and Systems Thinking." *Nature Chemistry* 8, no. 5: 393–398. <https://doi.org/10.1038/nchem.2498>.
- Müller, H., and K. Remke. 2025. "Circular Systems Sandbox: A Process Tool for Circular Systems Design and Rebound Effect Reduction." *Journal of Business Venturing Design* 4: 100028. <https://doi.org/10.1016/j.jbvd.2025.100028>.
- Muñoz, P., and O. Branzei. 2021. "Regenerative Organizations: Introduction to the Special Issue." *Organization & Environment* 34, no. 4: 507–516. <https://doi.org/10.1177/10860266211055740>.
- Muñoz, P., and B. Cohen. 2017. "Towards a Social-Ecological Understanding of Sustainable Venturing." *Journal of Business Venturing Insights* 7: 1–8. <https://doi.org/10.1016/j.jbvi.2016.12.001>.
- Muslemani, H., X. Liang, K. Kaesehage, F. Ascui, and J. Wilson. 2021. "Opportunities and Challenges for Decarbonizing Steel Production by Creating Markets for 'Green Steel' Products." *Journal of Cleaner Production* 315: 128127. <https://doi.org/10.1016/j.jclepro.2021.128127>.
- Neisig, M. 2025. "The Regenerative Semantics and Structural Change: Social Systems Using Nature as a Regenerative Medium." *Systems Research and Behavioral Science* 42, no. 2: 531–544. <https://doi.org/10.1002/sres.3146>.
- Ogunwole, O., E. C., Onukwulu, M. O., Joel, E. M., Adaga, and A. I. Ibeh. 2023. "Modernizing Legacy Systems: A Scalable Approach to Next-Generation Data Architectures and Seamless Integration." *International Journal of Multidisciplinary Research and Growth Evaluation* 4, no. 1: 901–909. <https://doi.org/10.54660/IJMRGE.2023.4.1.901-909>.
- Ostrom, E. 2007. "A Diagnostic Approach for Going Beyond Panaceas." *Proceedings of the National Academy of Sciences of the United States of America* 104, no. 39: 15181–15187. <https://doi.org/10.1073/pnas.0702288104>.
- Patagonia. 2021. "Annual Benefit Corporation Report Fiscal Year 2021." <https://www.patagonia.com/on/demandware.static/-/Library-Sites-PatagoniaShared/default/dw18ad9c7c/PDF-US/Patagonia-2021-BCorp-Report-Updated-2-15-22.pdf>.
- Paul, L., J. Väättänen, H. Bhimani, and A. Mention. 2025. "Leading Regenerative Transformation: Cognitive Frame Characteristics Shaping Leaders' Approaches to Organizational Development." *Business Strategy and the Environment* 34: bse.70153. <https://doi.org/10.1002/bse.70153>.
- Peffer, K., T. Tuunanen, M. A. Rothenberger, and S. Chatterjee. 2007. "A Design Science Research Methodology for Information Systems Research." *Journal of Management Information Systems* 24, no. 3: 45–77. <https://doi.org/10.2753/MIS0742-1222240302>.
- Pieroni, M. P. P., T. C. McAloone, and D. C. A. Pigosso. 2020. "From Theory to Practice: Systematising and Testing Business Model Archetypes for Circular Economy." *Resources, Conservation and Recycling* 162: 105029. <https://doi.org/10.1016/j.resconrec.2020.105029>.
- Rahman, S., N. T. Nguyen, and N. Slawinski. 2024. "Regenerating Place: Highlighting the Role of Ecological Knowledge." *Organization & Environment* 37, no. 3: 466–494. <https://doi.org/10.1177/10860266231220081>.
- Remke, K., and H. W. Müller. 2024. "Overcoming Rebound Effects: A Process Blueprint for Circular Systems Design." In *Lecture Notes in Computer Science*, 33–47. Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-61175-9_3.
- Rhodes, C. J. 2017. "The Imperative for Regenerative Agriculture." *Science Progress* 100, no. 1: 80–129. <https://doi.org/10.3184/003685017x14876775256165>.
- Salonen, K., P. Ritala, and N. Bocken. 2025. "Emerging Regenerative Business Paradigm: Narrative Review, Synthesis, and Research Agenda." *Journal of Circular Economy* 3, no. 3. <https://doi.org/10.55845/DOZQ3944>.
- Schaltegger, S., F. Lüdeke-Freund, and E. G. Hansen. 2016. "Business Models for Sustainability: A Co-Evolutionary Analysis of Sustainable Entrepreneurship, Innovation, and Transformation." *Organization & Environment* 29, no. 3: 264–289. <https://doi.org/10.1177/1086026616633272>.
- Schilling, L., and S. Seuring. 2024. "Linking the Digital and Sustainable Transformation With Supply Chain Practices." *International Journal of Production Research* 62, no. 3: 949–973. <https://doi.org/10.1080/00207543.2023.2173502>.
- Schreefel, L., R. P. O. Schulte, I. J. M. De Boer, A. P. Schrijver, and H. H. E. Van Zanten. 2020. "Regenerative Agriculture—The Soil is the Base." *Global Food Security* 26: 100404. <https://doi.org/10.1016/j.gfs.2020.100404>.
- Seckler, C., R. Mauer, and J. Vom Brocke. 2021. "Design Science in Entrepreneurship: Conceptual Foundations and Guiding Principles." *Journal of Business Venturing Design* 1, no. 1–2: 100004. <https://doi.org/10.1016/j.jbvd.2022.100004>.
- Seckler, C., R. Mauer, and J. Vom Brocke. 2023. "A Design Science Methodology for the Entrepreneurship Field." *Academy of Management Proceedings* 2023, no. 1: 13493. <https://doi.org/10.5465/amproc.2023.13493abstract>.
- Shannon, G., R. Issa, C. Wood, and I. Kelman. 2022. "Regenerative Economics for Planetary Health: A Scoping Review: Research Paper." *International Health Trends and Perspectives* 2, no. 3: 81–105. <https://doi.org/10.32920/ihtp.v2i3.1704>.
- Sheldon, R. A. 2017. "The E Factor 25 Years on: The Rise of Green Chemistry and Sustainability." *Green Chemistry* 19, no. 1: 18–43. <https://doi.org/10.1039/c6gc02157c>.
- Slawinski, N., B. Winsor, D. Mazutis, J. W. Schouten, and W. K. Smith. 2021. "Managing the Paradoxes of Place to Foster Regeneration." *Organization & Environment* 34, no. 4: 595–618. <https://doi.org/10.1177/1086026619837131>.
- Soderstrom, S. B., and K. Weber. 2020. "Organizational Structure From Interaction: Evidence From Corporate Sustainability Efforts." *Administrative Science Quarterly* 65, no. 1: 226–271. <https://doi.org/10.1177/0001839219836670>.
- Steffen, W., K. Richardson, J. Rockström, et al. 2015. "Planetary Boundaries: Guiding Human Development on a Changing Planet." *Science* 347, no. 6223: 1259855. <https://doi.org/10.1126/science.1259855>.
- Strauss, A. L., and J. M. Corbin. 2003. *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*. 2nd ed. Sage Publ.
- Trunk, J., K. Remke, and C. Seckler. 2025. "Trash to Treasure: Design Principles for Developing Cascading and Repurposing Business Models Through Systemic Stakeholder Collaboration." Hawaii International Conference on System Sciences (HICSS). <https://hdl.handle.net/10125/109452>.
- Tuunanen, T., R. Winter, and J. Vom Brocke. 2024. "Dealing With Complexity in Design Science Research: A Methodology Using Design Echelons." *MIS Quarterly* 48, no. 2: 427–458. <https://doi.org/10.25300/misq/2023/16700>.
- Van Aken, J. E., and H. Berends. 2018. *Problem Solving in Organizations: A Methodological Handbook for Business and Management Students*. 3rd ed. Cambridge University Press. <https://doi.org/10.1017/9781108236164>.
- Van Geem, K. M., V. V. Galvita, and G. B. Marin. 2019. "Making Chemicals With Electricity." *Science* 364, no. 6442: 734–735. <https://doi.org/10.1126/science.aax5179>.

Venable, J., J. Pries-Heje, and R. Baskerville. 2016. "FEDS: A Framework for Evaluation in Design Science Research." *European Journal of Information Systems* 25, no. 1: 77–89. <https://doi.org/10.1057/ejis.2014.36>.

Vlasov, M. 2021. "In Transition Toward the Ecocentric Entrepreneurship Nexus: How Nature Helps Entrepreneurs Make Ventures More Regenerative Over Time." *Organization & Environment* 34, no. 4: 559–580. <https://doi.org/10.1177/1086026619831448>.

Wesseling, J. H., E. M. M. I. Niesten, J. Faber, and M. P. Hekkert. 2015. "Business Strategies of Incumbents in the Market for Electric Vehicles: Opportunities and Incentives for Sustainable Innovation." *Business Strategy and the Environment* 24, no. 6: 518–531. <https://doi.org/10.1002/bse.1834>.

Wurlod, J.-D., and J. Noailly. 2018. "The Impact of Green Innovation on Energy Intensity: An Empirical Analysis for 14 Industrial Sectors in OECD Countries." *Energy Economics* 71: 47–61. <https://doi.org/10.1016/j.eneco.2017.12.012>.

Zhu, L., J. Luo, Q. Dong, Y. Zhao, Y. Wang, and Y. Wang. 2021. "Green Technology Innovation Efficiency of Energy-Intensive Industries in China From the Perspective of Shared Resources: Dynamic Change and Improvement Path." *Technological Forecasting and Social Change* 170: 120890. <https://doi.org/10.1016/j.techfore.2021.120890>.

Zimmerman, J. B., P. T. Anastas, H. C. Erythropel, and W. Leitner. 2020. "Designing for a Green Chemistry Future." *Science* 367, no. 6476: 397–400. <https://doi.org/10.1126/science.aay3060>.

Appendix

Interview Guide 1 – Chemist

Objective:

Understand the daily challenges and opportunities chemists face in the lab: Gathering detailed insights into the sustainability challenges and opportunities inherent in current laboratory practices.

Introduction

- Short explanation of the objective and purpose of the interview.
- Confidentiality and consent to record the session.

Background Questions

- Can you describe your typical day in the lab/working place?
- What are the primary objectives of your current research or projects?

Process and Materials

- How do you select the materials and chemicals for your experiments?
- Are there any materials you avoid due to environmental or sustainability concerns? Could you share any examples?

Waste and Efficiency

- What waste products result from your lab work?
- How do you manage or minimize waste in your processes?

Green Chemistry

- Are you familiar with the concept of green chemistry?
- Do you have examples of green chemistry innovation you have implemented or considered?

Regeneration

- Are you familiar with the concept of regeneration?
- What changes would you like to see in the chemical industry to make it more regenerative?

Conclusion

- Is there anything you would like to add that we have not discussed?

- Do you know anyone who could participate in this study?
- Would you like to follow up and stay updated on the research results?

Interview Guide 2—Entrepreneurs and Innovators in Chemical Industry

Objective

Explore the practical applications of chemical work in the business field and gather entrepreneurial insights: Understanding how regenerative practices can be applied in real-world settings and to explore the potential commercial viability and market acceptance of regenerative business models.

Introduction

- Short explanation of the objective and purpose of the interview.
- Confidentiality and consent to record the session.

Professional Background

- Could you describe your typical day in your workplace?
- What are the objectives and purpose of your company?

Business Model and Sustainability

- What are the core components of the business model of your company?
- How important is sustainability in your business model?

Challenges and Opportunities

- What are the biggest challenges in your job?
- The chemical industry is highly regulated, how important are the regulations in your field?
- Do you feel that there is an increase of awareness on the importance of sustainability in the industry?

Green Chemistry

- Are you familiar with the concept of green chemistry?
- Do you have examples of green chemistry innovation you have implemented or considered?

Regeneration

- Are you familiar with the concept of regeneration?
- What changes would you like to see in the chemical industry to make it more regenerative?

Conclusion

- Is there anything you would like to add that we have not discussed?
- Do you know anyone who could participate in this study?
- Would you like to follow up and stay updated on the research results?

Interview Guide 3—Sustainability Experts

Objective:

Gain insights into regenerative business principles, and best practices: Leveraging expert knowledge in the principles and successful applications of regenerative business models.

Introduction

- Short explanation of the objective and purpose of the interview.
- Confidentiality and consent to record the session.

Professional Background

- Could you describe your typical day in your workplace?

- What are the objectives and purpose of your company?

Expertise and Experience

- Can you describe your expertise in regenerative business models?
- What led you to specialize in this area?

Impact and Design

- How do you define a regenerative business model?
- What long-term impacts do regenerative business models have on industries and society?
- Which aspects of regenerative business model design do you find most critical?
- How would you measure the success of a regenerative business model?

Chemical Industry Application

- How can regenerative models be tailored to the chemical industry?

Barriers and Enablers

- What barriers typically could prevent chemical companies from adopting regenerative practices?
- What factors can enable a successful transition to a regenerative model?

Conclusion

- Is there anything you would like to add that we have not discussed?
- Do you know anyone who could participate in this study?
- Would you like to follow up and stay updated on the research results?